

Bulk Gauging and Quantum Couplings across a Compact Higgs Transition

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ABSTRACT: We study an explicit compact Calabi–Yau transition in which two conifold sectors and a rank-one E_2 sector can Higgs only together. The integral relation between their curve classes determines two shared Abelian gaugings and a transition from Hodge numbers $(6, 122)$ to $(3, 123)$. We compute the primitive surviving charge lattice and its cubic couplings, and show that the flavor gauging cannot remain dynamical in a gravity-decoupling limit with bounded local Kähler scales. On an auxiliary circle the quantum transition contains four conifold states spanning a primitive rank-three electric lattice. Their one-loop thresholds reproduce the integral conifold monodromies. An explicit Laurent mutation and analytic continuation of all eight regular periods identify the genus-zero prepotential of the surviving vectors with that of the smooth phase, including the integral normalization and constant term, and hence its scalar and electromagnetic couplings. At fixed five-dimensional Planck scale the auxiliary-circle limit retains finite bulk gauging, but its closing Kaluza–Klein gap prevents a uniform four-state description. The K3 and elliptic fibrations identify a dilaton gauging, an anomaly-consistent six-dimensional $U(2)$ phase and equal compact images of two E-string roots. An exact volume bound excludes the six-dimensional limit on the transition face. A heterotic K3 candidate and topological data are obtained; the global bundle construction and the finite-Planck hypermultiplet geometry remain open.

KEYWORDS: M-Theory, Supersymmetric Gauge Theory, Supergravity Models, String Duality, Differential and Algebraic Geometry

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1 Introduction

In a compact Calabi–Yau threefold, distinct singular regions can couple to the same vector fields. This changes which combinations of their local Higgs directions are available. At a conifold, the relevant degrees of freedom are charged wrapped-M2 hypermultiplets [1–3]. At a collapsing del Pezzo surface they instead form an interacting five-dimensional fixed point [4–7]. For a degeneration containing both kinds of sector, the compactification gauges flavor Cartans of the interacting theory together with the charges of the nodal hypermultiplets. The resulting constraints need not separate into independent local transitions.

We study this mechanism and the vector-multiplet couplings across one explicit compact transition. Its singular member X_9 has two ordinary double points and a cone over $\text{Bl}_2 \mathbb{P}^2$, which engineers the rank-one E_2 theory. In a crepant resolution $\widehat{X}_9$, let C_1, C_2 be the two exceptional nodal curves and let R be the image of the $SU(2)$ flavor root of the del Pezzo surface. The basic relation is integral:

$$R = C_1 + C_2 \quad \text{in } H_2(\widehat{X}_9; \mathbb{Z}). \quad (1.1)$$

The two independent compact gaugings therefore leave a single joint Higgs direction. No local sector can move alone. We use *cooperative Higgsing* for this situation: a global deformation activates several local sectors, but no nonzero deformation supported on any one of them is allowed. The explicit smoothing exists to all orders and changes the Hodge numbers as

$$(h^{1,1}, h^{2,1}) = (6, 122) \longrightarrow (3, 123). \quad (1.2)$$

Our first result identifies the compact charge constraint and its physical scope. Wrapped-M2 charges and flavor-root pairings define a common linear map Q . On a chosen local Higgs component, the image of the rigid triplet-moment-map quotient in invariant deformation coordinates is $\ker Q$. For the admissible toric hypersurfaces specified below, this is also the image of the global first-order deformation map. The exact mixed smoothing theorem further makes this a necessary and sufficient existence test after the local discriminants are excluded. Its selected-base integrability statement and the explicit construction of the smoothing are established in [8]; we recall their hypotheses and the local identifications. Thus existence of the X_9 transition is both a consequence of the general criterion and an explicit construction used to determine its smooth phase. The invariant image is not the full quaternionic-Kähler hypermultiplet manifold of the compact supergravity theory. In particular, a rigid quotient does not by itself determine its finite-Planck metric [9].

For X_9 , the rank-two constraint accounts for two Higgsed vectors; a third vector is lost when the E_2 Coulomb direction disappears. The surviving charge lattice is computed integrally, not only over $\mathbb{Q}$. Its cubic and curvature couplings agree with an independently constructed smooth threefold. Locally a contractible $\mathbb{F}_1$ is replaced by a rigid $\mathbb{P}^2$, giving the E_0 sector on the smooth side. This calculation also distinguishes the primitive unbroken torus from the cyclic group appearing in the formal invariant ring.

Our second result concerns decoupling gravity. We classify the nef directions in which the total volume diverges while the nodal masses and the entire del Pezzo Kähler class remain bounded. Both vectors imposing the joint constraint become non-dynamical in every such limit. One boundary direction can retain a different nodal gauging; no direction retains the E_2 flavor-root gauging. Thus this particular coupling of the local sectors requires finite Planck mass, under the stated local-scale hypotheses. The conclusion follows from the compact intersection form and includes a calculation with the full inverse kinetic matrix.

The third and most detailed result determines the genus-zero quantum vector geometry across the transition. On an auxiliary circle, the quantum transition is a locus with four ordinary conifold states spanning a primitive rank-three electric lattice. Their one-loop logarithmic thresholds reproduce the integral conifold monodromies and vanish in the surviving neutral directions. This fixes the singular terms but not the regular couplings. An explicit residue-preserving Laurent mutation relates its mirror to that of the smooth phase. We continue the periods to this locus, prove descent of the seven regular D0, D2 and D4 periods, and identify the eighth period in an integral topological D-brane frame. This fixes the reduced prepotential, including its quadratic ambiguity and constant term. It therefore determines the surviving scalar kinetic metric, gauge-kinetic and theta matrices, including graviphoton mixing, on the marked branch.

Two limits have different physical outcomes. Sending the Calabi–Yau volume to infinity at fixed eleven-dimensional Planck length, while keeping the local scales bounded, removes the shared flavor gauging. Decompactifying the auxiliary spacetime circle at fixed five-dimensional Planck scale instead reaches the contraction with finite bulk kinetic terms. In the latter limit the Kaluza–Klein gap closes, so the four-hypermultiplet description has no radius-independent energy range. The compact five-dimensional transition can thus retain the shared gauging without admitting a uniform four-particle description.

The fibration structures provide independent physical checks. One gauging vector is the K3-fiber class, identified with the heterotic dilaton multiplet. A flat elliptic phase has a six-dimensional $U(2)$ theory with anomaly-consistent matter and a geometrically determined height pairing. The relation (1.1) identifies the images of roots in two E-string lattices. An exact volume bound excludes the six-dimensional F-theory limit on the transition face. The heterotic instanton and five-brane counts and an explicit candidate K3 are determined, but the global bundle construction and its massless $U(1)$ remain open.

Relation to earlier work. The charged-hypermultiplet description of conifold transitions and their supergravity couplings is standard [2, 9]. Compact del Pezzo transitions, including examples involving additional conifold singularities, were studied in [10]. The local Higgs rings used here are those of [11]; the deformation rings follow from Altmann’s theory [12]. Conifold transitions relate variations of Hodge structure and quantum cohomology in a general framework [13], and the toric Frobenius method is well established [14, 15]. Our contribution is the explicit combination of a mixed compact charge constraint, its local-scale decoupling analysis, and an integral analytic period comparison for the same transition. We do not

propose a new general mirror algorithm or a general theorem that all such mixed transitions are smoothable.

Organization. Section 2 states the compact constraint and its supergravity qualification. Section 3 gives the transition and integral classical couplings, and Section 4 studies gravity-decoupling limits. Section 5 develops the quantum transition and period comparison. Section 6 gives the fibration interpretation. The appendices contain the deformation, lattice, analytic continuation and fibration calculations. Magnetic sheaves and their stability walls are treated separately in [16]; none of the period proofs below depends on that investigation.

2 Compact charge constraints

2.1 Wrapped-brane charges and deformation coordinates

Let $\pi : \widehat{X} \rightarrow X$ be a projective crepant resolution of a compact Calabi–Yau threefold with isolated singularities. In an integral divisor basis, the three-form reduction and wrapped-M2 coupling are

$$C_3 = \sum_I A^I \wedge \omega_I + \cdots, \quad q_I(C) = \int_C \omega_I = D_I C. \quad (2.1)$$

For a resolved node the exceptional curve supports the conifold hypermultiplet [2, 3, 9]. In one complex structure its scalars $(h, \tilde{h})$ have opposite charges, and the invariant deformation coordinate is

$$u = h\tilde{h}, \quad xy - zw = u. \quad (2.2)$$

For $u = |u|e^{i\theta}$, taking $(h, \tilde{h}) = \sqrt{|u|}(e^{i\theta}, 1)$ also sets the real moment map to zero.

The anticanonical cone over $S_7 = \text{Bl}_2 \mathbb{P}^2$ engineers the rank-one E_2 theory, with flavor algebra $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ [4, 5]. Its reduced Higgs ring is

$$\mathbb{C}[J_+, J_0, J_-]/(J_+J_- - J_0^2), \quad \beta = J_0. \quad (2.3)$$

The operators $J_\pm$ are the instanton currents in its $SU(2)$ gauge-theory deformation with one flavor [11]. The geometric deformation coordinate β is the Cartan invariant, not the whole two-complex-dimensional Higgs cone. Normalize its effective Cartan action to give J_+ charge one. On the double cover,

$$(J_+, J_0, J_-) = (a^2, ab, b^2)/2, \quad (a, b) \sim (-a, -b), \quad \mu^{\mathbb{R}} = \frac{1}{4}(|a|^2 - |b|^2). \quad (2.4)$$

Equal absolute values again lift every complex β at zero real moment map. The cover coordinates are not chiral-ring generators. The full ring also contains a nilpotent operator; its scheme-theoretic role is recorded in Appendix A.

For the root $R = e_1 - e_2$ on S_7 , the compact charge of the raising current is the difference of the two exceptional-curve charges:

$$q_I(e_1) - q_I(e_2) = D_I \iota_* R. \quad (2.5)$$

It is not twice this difference. In the standard $SU(2)$ Cartan normalization, where a root has weight two, the corresponding coefficient is $\frac{1}{2}D_{I^*}R$. The effective action on the reduced ring need not by itself extend to all states; Appendix B checks the extension in X_9 using the diagonal center identification with the other Abelian and Coulomb factors.

The degree-six del Pezzo cone similarly has E_3 flavor algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$. Its two reduced deformation components are a Cartan plane and a Cartan line, corresponding to the A_2 and A_1 root sublattices of $K_{S_6}^\perp$. A *branch profile* σ chooses one component at each such point. No E_3 point occurs in X_9 ; the branchwise formulation is useful for stating the general comparison precisely.

2.2 The rigid and geometric constraints

Let $\mathcal{B}_\sigma$ be the direct sum of the nodal lines, the reduced E_2 lines and the selected E_3 Cartan spaces. Choose compatible root-marked coordinates. The map

$$Q_{X,\sigma} : \mathcal{B}_\sigma \longrightarrow H^2(\widehat{X}; \mathbb{Q})^\vee \otimes \mathbb{C} \quad (2.6)$$

has columns the divisor pairings with exceptional nodal curves and the pushforwards of the selected roots. Since moment maps add under diagonal gauging, its components are the total complex moment maps of the rigid local-sector model [17].

Proposition 2.1. *The image of the zero-level rigid triplet-moment-map quotient in the invariant coordinates $\mathcal{B}_\sigma$ is $\ker Q_{X,\sigma}$.*

Proof. The complex moment-map equations give one inclusion. Conversely, lift each coordinate of a vector in the kernel by the balanced representatives above. For an E_3 component use the rank-one ADHM representative with $|q_i| = |\tilde{q}_i|$ and $\sum_i q_i \tilde{q}_i = 0$, as detailed in Appendix A. Each local real Cartan moment map vanishes separately, and the complex equations vanish by the kernel condition. Quotienting does not change these invariant coordinate values. $\square$

The same kernel has a geometric interpretation. We use the fan polytope convention: $\mathbb{P}_\Delta$ is defined by cones over faces of a reflexive $\Delta \subset \mathbb{Z}^4 \otimes \mathbb{R}$; the polar Δ° is its anticanonical Newton polytope. Call Δ admissible when every edge has lattice length one, its two-faces are unimodular triangles, unit squares or unit-edge pentagons/hexagons with one interior lattice point, and every dual edge of a singular two-face has length one. A Δ -regular anticanonical hypersurface then has exactly the isolated local germs just described. These hypotheses make the ambient divisor calculation detect the full rational kernel [18, 19].

Theorem 2.2. *Let X be such a hypersurface, with $\omega_X \cong \mathcal{O}_X$, $h^1(X, \mathcal{O}_X) = 0$, and a projective crepant resolution. Identifying each branch tangent space with its root-marked invariant coordinates, the image of the global first-order deformation space $\mathbb{T}_X^1$ satisfies*

$$\mathrm{im} \left(\mathbb{T}_X^1 \longrightarrow \bigoplus_p T_{X,p}^1 \right) \cap \mathcal{B}_\sigma = \ker Q_{X,\sigma}. \quad (2.7)$$

The proof in Appendix A identifies the local-to-global obstruction map with the pushforward of link classes. Combined with Proposition 2.1, it identifies two images in $\mathcal{B}_\sigma$; it does not identify the entire Higgs branch with a complex vector space.

The exact mixed smoothing theorem strengthens this tangent comparison [8, Theorems 4.5 and 4.7]. Let $\mathcal{D}_\sigma$ be the union of the local discriminants in $\mathcal{B}_\sigma$. Then

$$X \text{ smooths on } \sigma \iff \ker Q_{X,\sigma} \setminus \mathcal{D}_\sigma \neq \emptyset. \quad (2.8)$$

At every rank-one summand the coefficient must be nonzero; on an E_3 plane all three coefficients of its root-marked vector $\mu_1 e_1 + \mu_2 e_2 + \mu_3 e_3$, with $\sum_i \mu_i = 0$, must be nonzero. Necessity includes arbitrary analytic smoothing arcs. The selected deformation base is smooth if the relation space projects nontrivially to every selected E_3 summand, and every tangent on it integrates. Appendix A states the precise scope. For X_9 there is no E_3 hypothesis to check, and its one-dimensional kernel has a vector nonzero at all three germs. The explicit smoothing in Section 3 additionally identifies the smooth phase needed for the charge and period comparisons.

2.3 The finite-Planck qualification

The charges are exact compactification data, whereas Proposition 2.1 uses rigid hyperkähler cones. A compact supergravity hypermultiplet manifold is instead quaternionic Kähler. Consider a two-derivative description with Abelian hypermultiplet-isometry gaugings, neutral vector scalars and no tensor multiplets. Let P_I^r be the quaternionic moment maps, k_I^X the Killing vectors, and h^I the very-special coordinates normalized by $\kappa_{IJK} h^I h^J h^K = 6$. At constant scalars and zero field strengths, the fermion shifts give the fully supersymmetric Minkowski conditions [9, 17]

$$h^I P_I^r = 0, \quad h_x^I P_I^r = 0, \quad h^I k_I^X = 0. \quad (2.9)$$

At a regular point, h^I and its independent tangent vectors h_x^I span the full vector space: the cubic gradient annihilates the tangents but has nonzero pairing with h^I . Hence

$$P_I^r = 0 \text{ for all } I, r, \quad h^I k_I^X = 0. \quad (2.10)$$

Gravity does not replace these individual equations by only the graviphoton combination. On the relevant massless locus the hyperino condition can vanish for the transition-sector charges; the quaternionic moment-map equations remain.

The charge embedding alone does not construct the nonlinear map from P_I^r to the rigid coordinates or a smooth hypermultiplet action at the interacting E_2 point. Explicit conifold supergravity models already approximate this metric [9]. We therefore keep three statements separate: the rigid invariant-coordinate constraint, the compact geometric smoothing, and the exact vector-sector comparison. The first two do not supply the missing quaternionic dictionary.

3 The mixed transition in X_9

3.1 Geometry and the smoothing

Let $S_7 = \text{Bl}_{p_1, p_2} \mathbb{P}^2$ denote the del Pezzo surface of degree seven. The compact threefold X_9 has two ordinary double points and one cone over S_7 . Its crepant resolution $\widehat{X}_9$ is an anticanonical hypersurface in the toric fourfold with fan polytope

$$\begin{aligned} \Delta_9 = \text{conv}\{ & v_0 = (1, 0, 0, 0), \ v_1 = (0, 1, 0, 0), \ v_2 = (-1, -1, 0, 0), \\ & v_3 = (0, 0, 1, 0), \ v_4 = (0, 0, 0, 1), \ v_5 = (-4, -2, -1, 0), \\ & v_6 = (-4, -2, 0, -1), \ v_7 = (3, 1, 1, 1), \ v_8 = (2, 1, 1, 1)\}. \end{aligned} \quad (3.1)$$

The singular two-faces are the squares 2367, 2457 and the pentagon 34568. We use the square diagonals 67 and 24, and the star subdivision of the pentagon at $p = (-2, -1, 0, 0)$. The exceptional divisor $E = D_p$ is S_7 ; the two exceptional curves are C_1, C_2 . The remaining boundary point $(-1, 0, 0, 0)$ lies in a facet interior and its divisor misses the hypersurface. The chamber and integral intersection data are specified in Appendices B and C.

Write h, e_1, e_2 for the standard basis of $H_2(E; \mathbb{Z})$, with intersection form $\text{diag}(1, -1, -1)$ and $K_E = -3h + e_1 + e_2$. The compact root relation is

$$R := \iota_*(e_1 - e_2) = C_1 + C_2. \quad (3.2)$$

Neither the nodal curves nor the root can deform independently. In reduced local coordinates (u_1, u_2, β) the allowed direction is $(-1, -1, 1)$, as the charge calculation below shows.

All three coefficients are nonzero, so (2.8) proves the existence of a convergent smoothing. The reduced selected base is smooth by the integrability theorem; this is not an assertion about the full nonreduced E_2 base. Independently, an explicit ambient construction gives a smoothing $\mathcal{X} \rightarrow \Delta$ with smooth general fiber $X_{9,t}$ [8, Theorem 11.8]. The construction uses the facet $F = \{v_2, \dots, v_8\}$ and deformation degree $(0, 1, -1, -1)$. On its ordered edges

$$(25, 26, 27, 36, 37, 38, 45, 47, 48, 56, 78) \quad (3.3)$$

the dilation vector is $(0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0)$. It defines compatible Minkowski decompositions into two segments on each square and a segment plus a unimodular triangle on the pentagon. The Ilten–Vollmert deformation and its relative anticanonical family therefore smooth all three germs [8, 20, 21], with

$$(u_1, u_2, \beta) = t(-1, -1, 1). \quad (3.4)$$

The construction determines more than abstract existence: its general ambient fiber is toric, with the fan specified in [8, Section 11.3]. It identifies the smooth threefold used in the classical and mirror comparisons below.

The massless multiplets on the two smooth sides are

	$h^{1,1}$	$h^{2,1}$	χ	n_V	n_H^{neutral}
$\widehat{X}_9$	6	122	-232	5	123
$X_{9,t}$	3	123	-240	2	124

(3.5)

Here $n_V = h^{1,1} - 1$ excludes the graviphoton and $n_H^{\text{neutral}} = h^{2,1} + 1$ includes the universal hypermultiplet. The general transition formula of Appendix A gives the same change as the independent construction of the general fiber: two flavor vectors are Higgsed, the rank-one E_2 Coulomb vector disappears, and one new neutral hypermultiplet remains.

3.2 Integral charges and the joint constraint

The six divisors

$$\mathcal{D} = (D_2, D_5, D_6, D_7, D_8, E) \quad (3.6)$$

form the full free integral H^2 lattice, not merely a saturated toric quotient. An explicit unimodular pairing proves this in Appendix B. Their pairings with (C_1, C_2, R) are

$$Q_{\mathcal{D}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (3.7)$$

An integral change of vector basis isolates

$$V_1 = -D_6, \quad V_2 = D_5, \quad Q_{X_9}^{\text{eff}} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}. \quad (3.8)$$

Both nonzero Smith invariants are one. Thus

$$u_1 + \beta = u_2 + \beta = 0, \quad \ker Q_{X_9}^{\text{eff}} = \mathbb{C}(-1, -1, 1). \quad (3.9)$$

Every nonzero allowed deformation moves all three sectors. The exceptional-divisor vector is neutral under this flavor constraint: $E|_E = K_E$ and $K_E \cdot (e_1 - e_2) = 0$. Its loss in (3.5) is the loss of the local Coulomb direction, not a third row of the rank-two flavor matrix.

The reduced rigid current ring gives a useful refinement of this coordinate statement. If $J_{\pm}, J_0$ are the E_2 instanton currents and $u_i = h_i \tilde{h}_i$, its invariant generators can be chosen as

$$X = h_1 h_2 J_-, \quad Y = \tilde{h}_1 \tilde{h}_2 J_+, \quad XY = \beta^4. \quad (3.10)$$

Consequently the formal reduced quotient is $\mathbb{C}^2/\mathbb{Z}_4$. The integral extension of the flavor action and the current-ring calculation are given in Appendix B. This is neither a finite-Planck hypermultiplet metric nor an unbroken discrete gauge group at a generic Higgs vacuum.

3.3 Surviving vectors and classical couplings

Let Y be the common complement of small singular neighborhoods. The nodal Milnor fibers have $H^2 = 0$. For the S_7 cone the Milnor fiber is $\text{Bl}_{pt} \mathbb{P}^3 \setminus S_7$; its restriction to the link has image $R^{\perp}/\mathbb{Z}K_E$. Integral Mayer–Vietoris gluing therefore gives

$$H^2(X_{9,t}; \mathbb{Z})_{\text{free}} \simeq \ker(C_1, C_2, R)/\mathbb{Z}E. \quad (3.11)$$

A primitive basis of this quotient is

$$(A, B, \nu) = (D_2 + D_7, D_5 + D_6 - D_7, \nu), \quad D_0 = A + 4B - 2\nu, \quad D_1 = A + 2B - \nu, \quad (3.12)$$

where ν agrees with D_8 on Y . The older convenient basis (D_0, D_1, ν) has index two and will be used only when that particular comparison simplifies the geometry.

The surface $D_8 \cong \mathbb{F}_1$ meets E in its exceptional curve. In the smoothing it becomes a rigid plane of class ν and normal bundle $\mathcal{O}_{\mathbb{P}^2}(-3)$, hence a local E_0 geometry [5]. The cut-and-paste calculation in Appendix B gives

$$\nu^3 = 9 \quad \text{versus} \quad D_8^3 = 8, \quad c_2(X_{9,t}) \cdot \nu = -6 \quad \text{versus} \quad c_2(\widehat{X}_9) \cdot D_8 = -4. \quad (3.13)$$

All cubic entries involving D_0 or D_1 agree when these fixed representatives on Y are used. In particular,

$$\mathcal{F}(\alpha D_0 + \beta D_1) = \frac{83\alpha^3 + 147\alpha^2\beta + 81\alpha\beta^2 + 14\beta^3}{6} \quad (3.14)$$

on both sides, with $\mathcal{F} = J^3/6$ in five-dimensional units.

For the quantum comparison a different lift is required: it must annihilate all four electric charges, not just their root difference. In the primitive nef basis $H_1, \dots, H_6$ of (C.1), this lattice is

$$\Lambda_{\text{mag}}^0 = \{P : PC_1 = PC_2 = P\iota_*e_1 = P\iota_*e_2 = 0\} = \mathbb{Z}H_2 \oplus \mathbb{Z}H_3 \oplus \mathbb{Z}H_4. \quad (3.15)$$

The classes in (3.12) have lifts

$$\tilde{A} = D_2 + D_7, \quad \tilde{B} = D_5 + D_6 - D_7 + E, \quad \tilde{\nu} = D_8 + E, \quad (3.16)$$

whose columns in (H_2, H_3, H_4) form

$$U = \begin{pmatrix} 0 & -1 & -2 \\ 2 & -1 & -1 \\ -1 & 1 & 1 \end{pmatrix}, \quad \det U = -1. \quad (3.17)$$

Indeed $P \mapsto P + (P \cdot \iota_*e_2)E$ induces the integral isomorphism from (3.11) to (3.15). With these lifts all ten cubic entries and all three second-Chern pairings agree with the smooth side. In lexicographic order in (A, B, ν) they are

$$(-1, 1, 4, -1, -4, -6, 3, 4, 6, 9), \quad (14, 30, -6). \quad (3.18)$$

The equality for unconfined lifts and the jump in (3.13) compare different resolved representatives. The E shifts reconcile them. Section 5 determines whether this integral polynomial comparison extends to quantum couplings.

The integral lattice also screens the local E_0 line-charge group. A plane line has (A, B, ν) charges $(2, -2, -3)$, whereas the compact class D_1^2 has charges $(1, 7, 1)$. Thus the local $\mathbb{Z}_3$ electric one-form symmetry does not persist in the compact charge lattice [22, Sections 3.4 and 5.3]. This uses integral combinations of effective curves; it does not assert an irreducible BPS representative of D_1^2 .

3.4 A wrapped-M5 anomaly check

Wrap an M5-brane on $S_- = D_8 \cong \mathbb{F}_1$ and compare it with the string on $S_+ = \nu \cong \mathbb{P}^2$. This comparison fixes the representative D_8 , rather than its unconfined lift $D_8 + E$. The restrictions of (D_0, D_1, D_8) to $\mathbb{F}_1 = \text{Bl}_{pt} \mathbb{P}^2$ are $(0, h, -3h + e)$, with $h^2 = 1, e^2 = -1, he = 0$; on the plane they are $(0, h, -3h)$. The M5 tensor lattice consequently loses one negative direction [23].

In the inflow conventions of [24, eq. (2.21)],

$$I_4(S) = -\frac{1}{2}(SD_A D_B)F^A F^B - \frac{S^3 + \frac{1}{2}c_2(X)S}{6}c_2(N) + \frac{c_2(X)S}{48}p_1(T\Sigma), \quad (3.19)$$

where N denotes the transverse $SU(2)$ bundle. The two transverse coefficients are both -1 , while

$$I_4(S_+) - I_4(S_-) = -\frac{1}{2}(F^\nu)^2 - \frac{1}{24}p_1(T\Sigma). \quad (3.20)$$

The removed left-moving lattice boson has unit ν charge and accounts for both terms. The exceptional direction approaches e in the blowdown limit; it need not be anti-self-dual at every polarization. Thus worldsheet content and bulk Chern–Simons inflow change together. This protected anomaly check does not identify an interacting infrared central charge or count the Higgs condensates.

4 Gravity-decoupling limits

4.1 Dynamical vectors and local scales

At finite compact volume, every harmonic two-form has finite norm and supplies a dynamical vector in the reduction of C_3 [25]. At the contraction, the graviphoton direction pairs trivially with the exceptional curves, so the two nonzero rows of (3.8) belong to vector multiplets. A gravity-decoupling limit can instead turn these vectors into background flavor sources. Whether their constraint survives is therefore a question about the full kinetic form, not the charge matrix alone.

Work at fixed eleven-dimensional Planck length. For a divisor combination P the inverse coupling, up to one common normalization, is the Hodge norm

$$\mathcal{N}_P(J) = \int_{\widehat{X}_9} \omega_P \wedge *_J \omega_P = -JP^2 + \frac{(J^2 P)^2}{4\mathcal{V}}, \quad \mathcal{V} = \frac{J^3}{6}. \quad (4.1)$$

This follows from Lefschetz decomposition of harmonic $(1,1)$ -forms [26, 27]. A retained vector needs a finite nonzero norm in the local units. A mode of divergent norm has vanishing canonically normalized coupling to a fixed local current. Once the decoupled vector is omitted, its moment-map equation is not imposed on the limiting local theory. All vector mixing must be included in this test.

We consider Kähler paths with

$$\mathcal{V} \longrightarrow \infty, \quad JC_1, JC_2, J|_E \text{ bounded}. \quad (4.2)$$

The first condition decouples five-dimensional gravity; the others keep the local particle and surface scales bounded. These hypotheses are essential. The finite-volume circle limit of Section 5.7 is a different operation.

4.2 The allowed nef face and retained vectors

The linear space annihilating the local scales is

$$H = \{w : wC_1 = wC_2 = 0, w|_E = 0\} = \text{span}(D_0, D_1). \quad (4.3)$$

It is also the pullback of $H^2(X_9; \mathbb{Q})$ to the resolution.

Proposition 4.1. *The intersection of the nef cone with H is $\mathbb{R}_{\geq 0}D_0 + \mathbb{R}_{\geq 0}D_1$.*

Proof. Both generators are restrictions of nef toric divisors. The effective curves $D_1D_8|_{\widehat{X}_9}$ and $D_0D_2|_{\widehat{X}_9}$ have (D_0, D_1) pairings $(0, 1)$ and $(4, 0)$, respectively. Hence a nef combination $\alpha D_0 + \beta D_1$ must have $\alpha, \beta \geq 0$, proving the reverse inclusion. $\square$

Lemma 4.2. *For a path satisfying (4.2), each normalized accumulation direction is a nonzero element $w = \alpha D_0 + \beta D_1$ of this cone and has $w^3 > 0$. If $\mathcal{N}_P$ stays bounded along a subsequence with direction w , then $wP = 0$ in $H^4(\widehat{X}_9; \mathbb{R})$. Conversely, along $J = J_* + \Lambda w$, every $P \in \text{ann}(w)$ has $\mathcal{N}_P \rightarrow -J_*P^2$.*

Proof. The local pairings define a linear map with kernel H . Their boundedness bounds the transverse component of J , while divergent volume forces its H component to diverge. Its normalized limits lie in the nef face. The cubic (3.14) is positive on every nonzero ray there. Writing $J = s(w + o(1))$, $s \rightarrow \infty$, gives

$$\mathcal{N}_P = s \left[-wP^2 + \frac{3(w^2P)^2}{2w^3} \right] + o(s). \quad (4.4)$$

The Hodge index inequality, extended to nef w by continuity, is $(w^2P)^2 \geq w^3(wP^2)$. The bracket is thus at least $(w^2P)^2/(2w^3) \geq 0$. Boundedness forces $w^2P = wP^2 = 0$. The form wPQ is negative semidefinite on $w^2P = 0$; a vector of zero square lies in its radical. Since $w^3 > 0$, orthogonality also to w gives $wP = 0$ in H^4 . For the converse, both JP^2 and J^2P are independent of Λ when $wP = 0$, and the denominator in (4.1) diverges. $\square$

Theorem 4.3. *No gravity-decoupling path satisfying (4.2) retains the E_2 Cartan gauging of X_9 . In particular the norms of $V_1 = -D_6$ and $V_2 = D_5$ diverge. The maximal retained subspaces for each leading direction, realized on straight Kähler rays, have the following charge rows on (u_1, u_2, β) :*

w	$\text{ann}(w)$	nonzero charge row	rank Q_{dyn}
D_1	$\text{span}(D_2, E)$	$(1, -1, 0)$	1
D_0	$\text{span}(D_8, E)$	<i>none</i>	0
$\alpha D_0 + \beta D_1, \alpha\beta > 0$	$\text{span}(E)$	<i>none</i>	0.

(4.5)

Proof. The intersection form gives the annihilators in (4.5); the charge rows follow from (3.7). None pairs nontrivially with R . The lemma bounds the retained subspace of any path by the appropriate annihilator. Further divergent subleading scales may remove vectors, but cannot restore a root-charged one. For a straight ray take $J_* = (6, 18, 16, -11, -8, 7)$ in the basis $\mathcal{D}$. The limiting norm matrices are $\text{diag}(14, 11)$ on (D_2, E) for $w = D_1$, and $\begin{pmatrix} 5 & -1 \\ -1 & 11 \end{pmatrix}$ on (D_8, E) for $w = D_0$; on the interior ray the E norm is 11. They are positive definite and realize the listed subspaces. Neither D_5 nor D_6 belongs to an annihilator, so their norms cannot stay bounded on any diverging sequence satisfying the hypotheses. $\square$

Thus a local limit may retain $u_1 = u_2$, but never $u_1 = u_2 = -\beta$. Other compactifications can retain common compact surfaces and shared gauging after gravity decouples [28, 29]; the theorem is specific to the cone and local-scale conditions of X_9 .

4.3 Current exchange with vector mixing

The inverse kinetic matrix makes the loss of the interaction explicit. For $J = t^I D_I$ set $L_{IJ} = \kappa_{IJK} t^K$ and $v = Lt$. Polarization of (4.1) gives

$$G = -L + \frac{vv^\top}{4\mathcal{V}}, \quad G^{-1} = -L^{-1} + \frac{tt^\top}{2\mathcal{V}}. \quad (4.6)$$

Multiplication verifies the second identity using $t^\top Lt = 6\mathcal{V}$. Let C be the first two columns of (3.7). The exchange coefficient of the two corresponding currents is $\mathbf{H} = C^\top G^{-1} C$, including all six vector fields. On $J = J_* + \Lambda w$,

$$(1, 1)\mathbf{H}_w(1, 1)^\top = \begin{cases} 2/(3\Lambda) + O(\Lambda^{-2}), & w = D_0, \\ 1/\Lambda + O(\Lambda^{-2}), & w = D_1, \\ 2/(5\Lambda) + O(\Lambda^{-2}), & w = D_0 + D_1. \end{cases} \quad (4.7)$$

The full matrices are printed in Appendix B. For $w = D_1$ the difference current retains a finite coupling, but the shared root current still decouples. Removing the graviphoton subtracts $tt^\top/(3\mathcal{V})$ from G^{-1} ; its contribution to $\mathbf{H}$ is $O(\Lambda^{-3})$, leaving (4.7) unchanged.

At every finite Λ , $\mathbf{H}$ is positive definite, so $(Q^{\text{eff}}b)^\dagger \mathbf{H} (Q^{\text{eff}}b)$ has kernel $\mathbb{C}(-1, -1, 1)$. Its limiting matrix has the larger kernel $\{u_1 = u_2\}$ on the D_1 ray and $\mathbb{C}^3$ on the other two. The vacuum constraint of the limiting local theory therefore need not be the limit of the finite-coupling constraint. This quadratic expression only diagnoses the rigid moment maps: the chosen J_* leaves $JC_1 = JC_2 = 1$, so these rays are not themselves Higgs vacua. At finite Planck mass the full conditions of (2.10) are still required.

5 Quantum vector multiplets across the transition

Compactify the five-dimensional theory on an auxiliary spacetime circle. Mirror symmetry then gives access to its four-dimensional vector multiplets through periods of a compact Calabi–Yau threefold. We construct a four-node degeneration, fix its integral charges in the

large-volume frame, and identify the surviving eight periods with those of the smooth side of Section 3. All analytic identities below refer to a specified continuation branch with small nonzero spectator parameters. The final subsection takes the auxiliary radius to infinity at fixed M-theory volume.

5.1 A compact four-node degeneration

Use the primitive nef divisors H_a and algebraic mirror coordinates z_a specified in Appendix C. Their dual integral curve classes are β_a , with

$$C_2 = \beta_1, \quad C_1 = \beta_5, \quad \iota_* e_2 = \beta_6, \quad \iota_* e_1 = \beta_1 + \beta_5 + \beta_6. \quad (5.1)$$

The notation β_a for curve classes is distinct from the local moment-map coordinate β . Set

$$\Sigma : z_1 = z_5 = z_6 = 1, \quad (v, a, b) = (z_2, z_3, z_4). \quad (5.2)$$

In a smooth ambient chart $(x, y, r, s) \in (\mathbb{C}^*)^2 \times \mathbb{C}^2$ the full compact mirror equation restricts to

$$\mathcal{F} = (1 + xy)(1 + x^{-1} + vy^{-1} + r) + s(r(br - 1) + as) = 0. \quad (5.3)$$

This is not a truncation in a or b . Its four nodes lie at

$$s = 0, \quad y = -x^{-1}, \quad r \in \{0, b^{-1}\}, \quad vx^2 - (1 + r)x - 1 = 0. \quad (5.4)$$

For $bv \neq 0$ and $(1 + r)^2 + 4v \neq 0$, the Hessian determinants are nonzero. The two nodes at $r = 0$ approach the local E_2 doublet; the other pair approaches the original conifold sectors in their square charts as $b \rightarrow 0$. Appendix C constructs the charts and excludes additional singularities at an exact rational sample, giving an open four-node stratum.

The local smoothing covectors in the three normal parameters (z_1, z_5, z_6) are

$$(x^{-1}, x^{-1} + r, -1). \quad (5.5)$$

They have rank three. The closure of $s = 1 + xy = 0$ is a compact Weil divisor passing once through each node. Its complement of the node neighborhoods supplies a four-chain with boundary the four oriented vanishing spheres. Thus there is one relation with unit coefficients, and the smoothing rank excludes a second. Blowing up this divisor gives a projective small resolution.

The four vanishing cycles are mutually local, with rank-three span. Standard type-IIB conifold physics therefore supplies four massless hypermultiplets charged under three Abelian vectors [1, 2]. Their Higgsing gives $(\Delta n_V, \Delta n_H^{\text{neutral}}) = (-3, +1)$ and mirror Hodge numbers $(122, 6) \rightarrow (123, 3)$. This is the finite-radius particle description. The family identification below establishes more than this numerical agreement with the original smoothing.

5.2 The integral massless charges

Proposition 5.1. *On a marked sufficiently small nonzero (v, a, b) branch of Σ , the large-volume flat coordinates satisfy*

$$t_1 = t_5 = t_6 = 0. \quad (5.6)$$

In D6–D4–D2–D0 order, the four vanishing charges are

$$(0, 0, C_1, 0), \quad (0, 0, C_2, 0), \quad (0, 0, \iota_* e_1, 0), \quad (0, 0, \iota_* e_2, 0). \quad (5.7)$$

Their span V is a primitive rank-three isotropic sublattice. Its unique relation, in this order, is $(-1, -1, 1, -1)$.

Proof. At fixed spectator degree, summation over all three local degrees gives convergent expressions for the scalar period W_0 and the six electric periods $W_0 t_a$. A binomial convolution sets the three local logarithmic corrections to zero on Σ . The branch is continued without coordinate winding, fixing the integer logarithmic offsets. The estimates and coefficient identities are proved in Appendix C.

The seven electric periods are single-valued near the locus. Picard–Lefschetz monodromy around each individual node therefore excludes D6 and D4 components from its vanishing charge. The three spectator flat coordinates are independent along Σ , excluding spectator D2 components and a D0 shift. Unit-normalized node residues, matched to the exceptional-curve and nodal flop limits, fix the four primitive labels in (5.1). The first, fifth and sixth dual basis elements generate their span, proving saturation; the integral cohomology tests in the appendix exclude hidden torsion. Mutual locality follows either from their pure D2 charges or from their disjoint Milnor neighborhoods. $\square$

The massless charge matrix, after removing neutral vectors, is

$$Q_{\text{light}} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}. \quad (5.8)$$

Its three Smith invariants are one. The formal zero-level hyperkähler quotient is the same $\mathbb{C}^2/\mathbb{Z}_4$ surface as (3.10). Here all four columns have an actual simultaneous massless-particle interpretation at finite radius. This does not determine a compact quaternionic-Kähler metric.

At a generic Higgs vacuum, a magnetic divisor charge P has oriented flux vector

$$(-PC_1, -PC_2, P\iota_* e_1, -P\iota_* e_2).$$

Its image is all of $\{w \in \mathbb{Z}^4 : \sum_i w_i = 0\}$, since the three independent electric pairings are surjective onto $\mathbb{Z}^3$. This is an A_3 winding lattice, not an $SU(4)$ enhancement. Its kernel is exactly (3.15); these are the magnetic directions whose periods can descend through the transition. The saturation also implies a connected unbroken gauge torus.

5.3 Hypermultiplet thresholds and their monodromy

The four charges determine the singular gauge thresholds without using the regular-period calculation. Near a generic point of Σ , choose transverse special coordinates $x = (t_5, t_1, t_6)$ and let q_α , $\alpha = 1, \dots, 4$, be the columns of (5.8). In the normalization $X^0 = 1$ their holomorphic central charges are

$$(Z_1, Z_2, Z_3, Z_4) = (x_1, x_2, x_1 + x_2 + x_3, x_3), \quad Z_\alpha = q_\alpha^\top x. \quad (5.9)$$

At fixed finite auxiliary radius, work off the discriminant and at energies below these four masses, the Kaluza–Klein scale and the other massive thresholds. Integrating out the four charged hypermultiplets gives the universal one-loop singularity of the holomorphic prepotential [1, Sections 2 and 4]:

$$F = F_{\text{reg}} + F_{\text{sing}}, \quad F_{\text{sing}} = \frac{1}{4\pi i} \sum_{\alpha=1}^4 Z_\alpha^2 \left(\log \frac{Z_\alpha}{z_*} - \frac{3}{2} \right). \quad (5.10)$$

Here $z_* \neq 0$ is a dimensionless reference scale, the logarithms are continued on the chosen branch, and F_{reg} is holomorphic across Σ . Changing z_* or the displayed quadratic subtraction only redistributes holomorphic quadratic terms between the two pieces; it does not change their sum in the fixed integral frame.

Twice differentiating gives the singular holomorphic coupling matrix

$$\mathcal{T}_{ij}^{\text{sing}} = \frac{\partial^2 F_{\text{sing}}}{\partial x_i \partial x_j} = \frac{1}{2\pi i} \sum_{\alpha=1}^4 q_{\alpha i} q_{\alpha j} \log \frac{Z_\alpha}{z_*}. \quad (5.11)$$

A positive loop around the α th discriminant therefore gives

$$\Delta_\alpha(\partial_x F) = q_\alpha Z_\alpha, \quad \Delta_\alpha \mathcal{T} = q_\alpha q_\alpha^\top. \quad (5.12)$$

These are precisely the unit Picard–Lefschetz shifts of the marked electric charges. If $x = \epsilon \xi$ with every $q_\alpha^\top \xi \neq 0$, winding ϵ once winds all four central charges. The combined shift is

$$\mathbf{B} = Q_{\text{light}} Q_{\text{light}}^\top = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad \mathcal{T}^{\text{sing}} = \frac{\log \epsilon}{2\pi i} \mathbf{B} + O(1). \quad (5.13)$$

Thus the logarithmic coefficient has rank three, although four hypermultiplets contribute. The Smith invariants of $\mathbf{B}$ are $(1, 1, 4)$, whereas those of Q_{light} are $(1, 1, 1)$. The index-four image of the combined monodromy is not the primitive vanishing lattice V and does not imply a residual discrete gauge group. Conifold reduction uses $V^\perp/V$.

The light charges have no components along the surviving coordinates (t_2, t_3, t_4) . Hence the singular terms have zero tangential derivatives on Σ : there is no logarithmic threshold for the unbroken neutral vectors. This does not determine their regular couplings. The remaining subsections identify those couplings with the smooth phase by continuing the full reduced period vector. At $Z_\alpha = 0$ the integrated-out action itself is inapplicable; the light hypermultiplets must be restored before Higgsing. Moreover, $\mathcal{T}$ is a holomorphic Hessian, not the full electromagnetic coupling matrix of compact supergravity, reconstructed below.

5.4 The small resolution as a smooth-side mirror

The holomorphic families can also be compared explicitly. The general ambient fiber of the smoothing of X_9 is toric: its fan, reconstructed in [8, Section 11], has primitive rays

$$\begin{aligned} r_0 &= (-1, -3, -1, 0), & r_1 &= (-1, -3, 0, -1), \\ r_2 &= (-1, -1, 0, 0), & r_3 &= (-1, 0, 0, 0), \\ r_4 &= (0, 1, 0, 0), & r_5 &= (1, -1, 0, 0), & r_6 &= (1, 3, 1, 1). \end{aligned} \tag{5.14}$$

Its maximal cones are 1456, 1346, 0456, 0346, 0145, 0134, 01256, 1236, 0236, 0123. This is a Gorenstein Fano fourfold; a general anticanonical hypersurface is the smooth fiber $X_{9,t}$. Consequently $P = \text{conv}(r_0, \dots, r_6)$ is the Newton polytope of its Batyrev mirror [18].

Proposition 5.2. *For generic (v, a, b) , the projective small resolution of the four-node mirror on Σ is birational to a smooth crepant Batyrev mirror of $X_{9,t}$. The birational map preserves the residue three-form, and the two smooth projective models are related by flops. The parameter family (v, a, b) has generically full rank in the three-dimensional complex-structure moduli space.*

Proof. Write (5.3) as $\mathcal{F} = fg + sk$, with $f = 1 + xy$, $g = 1 + x^{-1} + vy^{-1} + r$ and $k = r(br - 1) + as$. Blow up the compact Weil divisor given in this chart by (s, f) . In the affine blowup chart $s = Tf$, the strict transform is

$$g + Tr(br - 1) + aT^2f = 0. \tag{5.15}$$

On its dense torus set $u = r$, $p = x/r^2$ and $q = y/r^2$. Dividing (5.15) by the torus unit uT gives

$$F_{\text{sm}} = \frac{1}{Tu^3p} + \frac{v}{Tu^3q} + \frac{1}{Tu} + \frac{1}{T} + bu + \frac{aT}{u} + aTu^3pq - 1 = 0. \tag{5.16}$$

The exponents of its seven nonconstant monomials, in the order shown, are precisely (5.14). The inverse map on the dense open set $1 + u^4pq \neq 0$ is

$$x = u^2p, \quad y = u^2q, \quad r = u, \quad s = T(1 + u^4pq). \tag{5.17}$$

Thus this is a birational map from the actual small resolution, not just a comparison of Newton-polytope invariants.

The three-form is preserved as well. Put $h = s/r$ and $G = 1 + u^4pq$. In the integral old torus coordinates (h, u, p, q) , the original Laurent equation is $\mathcal{F}/(rs) = 0$. The substitution $h = TG/u$ changes this Laurent polynomial exactly to F_{sm} . Moreover

$$d \log h \wedge d \log u \wedge d \log p \wedge d \log q = d \log T \wedge d \log u \wedge d \log p \wedge d \log q, \tag{5.18}$$

because $d \log G$ involves only u, p, q . Taking residues proves the assertion, up to the common orientation choice. This is a Laurent mutation in the usual volume-preserving sense [30, Definition 1.1].

We must check that the coefficient restrictions in (5.16) do not impose an extra specialization. The lattice points of P are its seven vertices, the origin and $e_* = (0, -1, 0, 0)$. The point e_* lies in the relative interior of the facet with inward normal $n_* = (0, 1, -2, -2)$. It pairs nonnegatively with all other primitive facet normals. The corresponding toric root automorphism acts on anticanonical monomials by

$$\chi^m \mapsto \chi^m (1 + \lambda/u)^{1 + \langle n_*, m \rangle}. \quad (5.19)$$

At the coefficients in (5.16), an added coefficient c/u transforms to $(c - \lambda + b\lambda^2)/u$; the constant becomes $-1 + 2b\lambda$ and the coefficient of $(Tu)^{-1}$ becomes $1 + \lambda$. The derivative of the first coefficient with respect to λ is -1 at zero. Thus $c = 0$ is locally a slice for this automorphism. Normalize the constant coefficient to -1 . The four torus normalizations $c_{r_0} = c_{r_2} = c_{r_3} = 1$ and $c_{r_6} = c_{r_5}$ have exponent matrix with rows $(r_0, r_2, r_3, r_6 - r_5)$ and determinant 1. They leave exactly $(c_{r_1}, c_{r_4}, c_{r_5}) = (v, b, a)$, with no finite torus ambiguity.

For completeness, P and its polar have 9 and 162 lattice points; the sums of facet-interior lattice counts are 1 and 34, and both codimension-two corrections vanish. Batyrev's formulas give $(h^{1,1}, h^{2,1}) = (123, 3)$ and show that all three complex-structure deformations are polynomial. At $(v, a, b) = (1/2, 1/100, 1/10)$ the face polynomials are nondegenerate, including on the full torus; the exact check tests their saturated critical ideals. Nondegeneracy is open, so a maximal projective crepant partial toric resolution gives a smooth projective Calabi–Yau threefold generically [18, Section 4.5]. The small resolution constructed above is also smooth projective and Calabi–Yau on the generic four-node family. The two models are birational; hence they are connected by flops [31, Theorem 1]. $\square$

The fundamental period. An independent all-orders comparison is available. The three nonnegative primitive relations among (5.14) are

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 4 & 1 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}. \quad (5.20)$$

They generate every nonnegative integral relation: its coefficients satisfy $n_0 = n_1 = n_6 = \alpha$, $n_2 = \beta$, $n_3 = \gamma$, $n_4 = 4\alpha + 2\beta + \gamma$ and $n_5 = \alpha + \beta + \gamma$. With the constant term normalized as in (5.16), take

$$(\xi_1, \xi_2, \xi_3) = (va^2b^4, ab^2, ab), \quad (v, a, b) = (\xi_1/\xi_2^2, \xi_3^2/\xi_2, \xi_2/\xi_3). \quad (5.21)$$

The integral exponent matrix has determinant -1 . The normalized constant-term period is

$$W_{\text{sm}} = \sum_{\alpha, \beta, \gamma \geq 0} \frac{(8\alpha + 4\beta + 3\gamma)! \xi_1^\alpha \xi_2^\beta \xi_3^\gamma}{(\alpha!)^3 \beta! \gamma! (4\alpha + 2\beta + \gamma)! (\alpha + \beta + \gamma)!}. \quad (5.22)$$

The substitution $(\ell, m, n) = (\alpha, 2\alpha + \beta + \gamma, 4\alpha + 2\beta + \gamma)$ identifies every coefficient with (C.33). Its inverse is $(\alpha, \beta, \gamma) = (\ell, n - m - 2\ell, 2m - n)$; nonnegativity is exactly the support condition in that formula. Therefore $W_{\text{sm}} = W_0|_\Sigma$ to all orders, and analytically in their common small-bulk domain of convergence.

5.5 Quantum couplings of the surviving vectors

The smooth-side mirror determines protected vector-multiplet data beyond the cubic and second-Chern pairings. We compute its genus-zero Gopakumar–Vafa invariants and compare its electric and neutral magnetic periods with the continued periods on Σ . Conifold surgery and large-volume asymptotics identify these seven periods to all orders on the marked branch. The finite coefficient calculation provides an independent check of this identification.

The smooth-side chamber. Use the primitive surviving divisor basis $K_1 = H_2$, $K_2 = H_3$, $K_3 = H_4$, with curve degrees $\mathbf{d} = (\ell, m, n)$ defined by their three pairings. Rewriting (5.20) in the coordinates (v, a, b) gives

$$Q^{\text{sm}} = \begin{pmatrix} 1 & 1 & -2 & 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & 0 & 0 \end{pmatrix}, \quad Q_*^{\text{sm}} = (0, 2, 1)^T. \quad (5.23)$$

Its columns are ordered by $r_0, \dots, r_6$ and give the toric divisor classes in the K basis. The only nonsimplicial maximal cone is 01256, with circuit $r_0 + r_1 + r_6 = 2r_2 + r_5$. Its two simplicial subdivisions replace it, respectively, by

$$\{1256, 0256, 0125\}, \quad \{0156, 0126\}. \quad (5.24)$$

These modifications occur over the ambient fixed point avoided by the general anticanonical hypersurface. They therefore leave $X_{9,t}$ unchanged. Their ambient Mori cones, in degree coordinates, are

$$\begin{aligned} M_+ &= \text{cone}\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}, \\ M_- &= \text{cone}\{(-1, 0, 0), (1, 1, 0), (0, 0, 1)\}, \\ \overline{\text{NE}}(X_{9,t}) &\subseteq M_+ \cap M_- = \text{cone}\{(1, 1, 0), (0, 1, 0), (0, 0, 1)\}. \end{aligned} \quad (5.25)$$

No equality with the actual Calabi–Yau Mori cone is required. In either subdivision the ten cubic entries, ordered lexicographically by triples of K indices, and the second-Chern row are

$$(0, 2, 3, 6, 11, 17, 14, 27, 49, 83), \quad (36, 80, 146). \quad (5.26)$$

They coincide with the resolved intersections of H_2, H_3, H_4 . This fixes the smooth-side large-volume chamber used for the calculation. It does not select a crepant model of the birational mirror small resolution.

The period comparison. For the charge matrix (5.23), the Frobenius coefficient is

$$c_{\text{sm}}(\ell, m, n) = \frac{\Gamma(1 + 2m + n)}{\Gamma(1 + \ell)^3 \Gamma(1 + n - m - 2\ell) \Gamma(1 + 2m - n) \Gamma(1 + n) \Gamma(1 + m - \ell)}. \quad (5.27)$$

Its rational derivatives mean derivatives of $c_{\text{sm}}(\mathbf{d} + \boldsymbol{\rho})/c_{\text{sm}}(\boldsymbol{\rho})$ at zero, as in (C.3). In particular, zeros from reciprocal Gamma factors must be retained before differentiation. Write

$W_{\text{sm}}, C_i^{\text{sm}}, \widehat{C}_{ij}^{\text{sm}}$ for the resulting scalar, first- and second-derivative series, and put

$$D_i^{\text{sm}} = \frac{1}{2} \sum_{j,k} \kappa_{ijk}^{\text{sm}} \widehat{C}_{jk}^{\text{sm}}, \quad g_i^{\text{sm}} = C_i^{\text{sm}} / W_{\text{sm}}. \quad (5.28)$$

The large-volume mirror construction then applies with (5.26) [15, Section 4.4].

Proposition 5.3. *On the marked sufficiently small, nonzero (v, a, b) branch of Σ , the scalar, spectator electric and contracted neutral magnetic series satisfy the analytic identities*

$$\begin{aligned} W_{\text{sm}} &= W_0|_{\Sigma}, \\ C_i^{\text{sm}} &= C_{i+1}|_{\Sigma}, \quad D_i^{\text{sm}} = D_{H_{i+1}}|_{\Sigma}, \quad i = 1, 2, 3. \end{aligned} \quad (5.29)$$

Consequently $\tau_i = t_{i+1}|_{\Sigma}$ and

$$\frac{\partial F_{\text{inst}}^{\text{sm}}}{\partial \tau_i} = \Psi_{H_{i+1}}|_{\Sigma}. \quad (5.30)$$

Here $F_{\text{inst}}^{\text{sm}}$ is the nonpolynomial genus-zero prepotential defined in (5.36). The statement concerns the specified continuation, not transport around arbitrary loops in moduli space.

Proof. Let $\check{X}$ be a nearby smooth member of the original mirror, and $\check{Y}$ its projective small resolution at the four-node locus. The vanishing-cycle space $V \subset H_3(\check{X}, \mathbb{Q})$ is isotropic of rank three by the integral marking. Conifold surgery identifies the surviving variation of Hodge structure with

$$H_3(\check{Y}, \mathbb{Q}) \simeq V^{\perp}/V, \quad \dim(V^{\perp}/V) = 14 - 2 \cdot 3 = 8. \quad (5.31)$$

Periods of classes in $V^{\perp}$ restrict to the corresponding periods of $\check{Y}$; their values on V vanish. This is the weight-three quotient in the limiting mixed Hodge structure [13, Corollary 1.11 and Section 3.3]. All electric classes and the magnetic classes with divisor part H_2, H_3, H_4 lie in $V^{\perp}$: the vanishing charges are pure D2 charges, and these three divisors annihilate them.

Proposition 5.2 identifies $\check{Y}$ birationally with the smooth-side Batyrev mirror and preserves the residue form. The two smooth models are connected by flops. Each flop is an isomorphism away from curves; three-cycles can be moved off those curves, so their period integrals are preserved. Applied over a local parameter neighborhood, this identifies their period spaces with the Gauss–Manin connection. Thus the restricted neutral periods and the smooth-side Frobenius periods belong to the same eight-dimensional space of local period functions. The scalar identity fixes their common normalization of the form.

It remains to identify the particular functions within that space. This step uses unnormalized periods, not the nonlinear regular corrections Ψ_P . Put $\lambda = (\log v, \log a, \log b)/(2\pi i)$. For either collection of series (W, C_i, D_i) , define

$$\begin{aligned} \Pi_i^{\text{el}} &= W\lambda_i + \frac{C_i}{2\pi i}, \\ \Pi_i^{\text{mag}} &= \frac{W}{2} \sum_{j,k} \kappa_{ijk}^{\text{sm}} \lambda_j \lambda_k + \frac{1}{2\pi i} \sum_{j,k} \kappa_{ijk}^{\text{sm}} \lambda_j C_k + \frac{D_i}{(2\pi i)^2}. \end{aligned} \quad (5.32)$$

These are linear combinations of Frobenius periods. Subtracting the universal $\zeta(2)$ term in the rational second derivative only adds a constant multiple of W . For the resolved periods restricted to Σ , the local logarithms and the three local C_a vanish. Their remaining cubic contractions are exactly (5.26), giving (5.32) with $C_i = C_{i+1}|_\Sigma$ and $D_i = D_{H_{i+1}}|_\Sigma$.

The constant terms of both collections are

$$W(0) = 1, \quad C_i(0) = D_i(0) = 0. \quad (5.33)$$

On the resolved side, every unbounded second-derivative sector is multiplied by a zero neutral pairing. All remaining local indices at spectator degree zero are zero, where the rational Gamma coefficient is identically one. This proves the last equality without truncating an infinite tail. The previously proved convergence and the smooth-side Frobenius expansion give the common polynomial asymptotics

$$W \sim 1, \quad \Pi_i^{\text{el}} \sim \lambda_i, \quad \Pi_i^{\text{mag}} \sim \frac{1}{2} \kappa_{ijk}^{\text{sm}} \lambda_j \lambda_k. \quad (5.34)$$

For example, on $(v, a, b) = (e^{-sw_1}, e^{-sw_2}, e^{-sw_3})$ with every $w_i > 0$, the omitted terms are bounded by $O(e^{-s \min w_i} (1+s)^2)$. All three weights may vary; this is a large-volume comparison at the origin, not the circle path with v fixed.

The smooth-side period space has a Frobenius basis with polynomial terms of degrees 0, 1, 1, 1, 2, 2, 2, 3 [15, Equations (4.15)–(4.19)]. Its three leading quadratics are those in (5.34). They are independent, as witnessed by

$$\det \begin{pmatrix} 0 & 2 & 3 \\ 2 & 6 & 11 \\ 3 & 11 & 17 \end{pmatrix} = 10. \quad (5.35)$$

The rows are $(\kappa_{i11}, \kappa_{i12}, \kappa_{i13})$. The eighth period has nonzero leading cubic $\kappa_{ijk}^{\text{sm}} \lambda_i \lambda_j \lambda_k / 6$, whose λ_3^3 coefficient is $83/6$. Therefore the eight polynomial asymptotics are linearly independent. No nonzero constant linear combination of these periods can be exponentially small for every positive weight vector: successively its cubic, quadratic, linear and constant polynomial coefficients would have to vanish.

Subtract the smooth-side functions in (5.32) from their restricted resolved counterparts. Each difference is a period function in this common space and has zero polynomial asymptotic by (5.33)–(5.34). Each difference is consequently zero. The electric identities first give equality of the C_i , and the magnetic identities then give equality of the D_i . The master equation (C.5) proves (5.30). $\square$

An independent coefficient check sums the resolved ten-monomial series over every local degree for all 35 triples with $\ell + m + n \leq 4$, and verifies (5.29) in exact arithmetic. The four unbounded second-derivative sectors of (C.40) are multiplied by $K_i \cdot C_1$, $K_i \cdot C_2$, $K_i \cdot \iota_* e_1$ or $K_i \cdot \iota_* e_2$, all zero. Every other nonzero sector has local indices at most $4(\ell + m + n)$, so each tested neutral coefficient includes its complete local-degree sum. The finite check does not

replace the period-space argument. The proposition identifies the three instanton gradients, and hence all genus-zero Yukawa couplings after the already established cubic match. This argument leaves a moduli-independent prepotential constant. Section 5.6 fixes it and the remaining D6 period using the polarization and the reality of the marked continuation, with the curvature and flux shifts of an integral symplectic basis. No equality of all uncontracted second Gamma derivatives is asserted.

Protected curve invariants. Define $q_i = \exp(2\pi i \tau_i)$ by $2\pi i \tau_i = \log(v, a, b)_i + g_i^{\text{sm}}$. The nonpolynomial genus-zero prepotential is

$$F_{\text{inst}}^{\text{sm}} = -\frac{1}{(2\pi i)^3} \sum_{\mathbf{d} \neq 0} n_{\mathbf{d}}^{0, \text{sm}} \text{Li}_3(q^{\mathbf{d}}). \quad (5.36)$$

Applying the three magnetic equations (C.5) with Q^{sm} determines $n_{\mathbf{d}}^{0, \text{sm}}$ for all 164 nonzero nonnegative degree vectors with $\ell + m + n \leq 8$. All three equations agree, every invariant is integral, and 31 are nonzero. The complete nonzero list through degree three is

$$\frac{\mathbf{d}}{n_{\mathbf{d}}^{0, \text{sm}}} \left| \begin{array}{ccccc} (0, 0, 1) & (1, 1, 0) & (0, 1, 1) & (1, 1, 1) & (0, 1, 2) \\ 10 & 3 & 140 & -20 & 140 \end{array} \right. \quad (5.37)$$

The degree is taken in the primitive K basis, not in the nonnegative-relation coordinates ξ_i .

The plane divisor is the r_2 column, $\nu = -2K_1 - K_2 + K_3$. Its square pairs as $(-3, -3, 0)$, so a plane line has primitive degree $(1, 1, 0)$. Along this ray the compact calculation gives

$$n_{(d, d, 0)}^{0, \text{sm}} = 3, -6, 27, -192, \quad d = 1, 2, 3, 4. \quad (5.38)$$

An independent one-parameter calculation with local $\mathcal{O}_{\mathbb{P}^2}(-3)$ charges $(1, 1, 1, -3)$ reproduces both these invariants and the corresponding coefficients of $g_1^{\text{sm}} + g_2^{\text{sm}}$. These are genus-zero protected M2 indices on the smooth side [32]; they are not unweighted state counts, magnetic probe indices, or the spectrum of the cooperative condensate at the transition.

5.6 The D6 period and the prepotential constant

The symplectic pairing and the reality of the marked continuation fix the eighth period. Curvature and flux shifts are essential in making this an integral statement. We first specify the D-brane basis; the D6 label below denotes a six-brane wrapping the threefold, not the toric divisor D_6 .

The integral symplectic frame. Appendix B constructs the primitive topological charge generators $\mathbf{m}_0, \mathbf{m}_a, \mathbf{e}_0, \mathbf{e}_a$, with pairing minus the Euler form. Use the ordered divisor bases $L_a = H_a$ on the resolution and $L_a = K_a$ on the smooth side, and write $c_a = c_2(X)L_a$, $\kappa_{abc} = L_a L_b L_c$.

Define the integers

$$I_a = \frac{\kappa_{aaa}}{6} + \frac{c_a}{12}, \quad h_{ab} = \frac{\kappa_{abb} - \kappa_{aab}}{2}. \quad (5.39)$$

Riemann–Roch gives $I_a \in \mathbb{Z}$ and $h_{ab} \in \mathbb{Z}$. The unimodular correction

$$\mathbf{m}'_0 = \mathbf{m}_0, \quad \mathbf{m}'_a = \mathbf{m}_a + I_a \mathbf{e}_0 + \sum_{b>a} h_{ab} \mathbf{e}_b \quad (5.40)$$

puts the pairing in standard symplectic form, leaving the electric generators fixed [15, Equations (2.20)–(2.24)]. The two index rows are $(2, 3, 9, 26, 36, 50)$ and $(3, 9, 26)$. In this convention the prepotential and normalized periods are

$$F^X(t) = -\frac{1}{6}\kappa_{abc}t_a t_b t_c + \frac{1}{2}a_{ab}t_a t_b + \frac{1}{24}c_a t_a + \frac{\chi(X)\zeta(3)}{2(2\pi i)^3} + F_{\text{inst}}^X(t), \quad (5.41)$$

$$G^X = 2F^X - t_a F_a^X, \quad F_a^X = \partial_{t_a} F^X, \quad a_{ab} = \begin{cases} \kappa_{aab}/2, & a \geq b, \\ \kappa_{abb}/2, & a < b. \end{cases}$$

Repeated indices are summed except in the definition of a_{ab} . The period G^X belongs to $\mathbf{m}_0$, which was not changed by (5.40). Equivalently, in the *unsubtracted* Frobenius derivatives,

$$W_X G^X = \frac{1}{6}\kappa_{abc}\varpi_X^{abc} + \frac{1}{12}c_a \varpi_X^a, \quad (5.42)$$

where ϖ_X^{abc} includes the factors $(2\pi i)^{-3}$ [15, Equations (2.29) and (4.19)]. Removing the $\zeta(3)$ term would change this marked period.

The surviving block of a_{ab} on the resolution equals the smooth-side matrix, just as the cubic and c_2 rows agree:

$$(a_{i+1,j+1}^{\text{res}})_{i,j=1}^3 = a^{\text{sm}} = \begin{pmatrix} 0 & 3 & 17/2 \\ 3 & 7 & 49/2 \\ 17/2 & 49/2 & 83/2 \end{pmatrix}. \quad (5.43)$$

Proposition 5.3 therefore identifies the seven unnormalized integral periods belonging to $\mathbf{e}_0, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4, \mathbf{m}'_2, \mathbf{m}'_3, \mathbf{m}'_4$ with their smooth-side counterparts. For example their magnetic components are

$$W F_i = -\Pi_i^{\text{mag}} + \frac{c_i}{24} W + \sum_j a_{ij} \Pi_j^{\text{el}}. \quad (5.44)$$

Local electric terms on the resolution vanish on Σ .

Proposition 5.4. *On the marked small-bulk branch of Σ , the reduced integral symplectic period vector agrees with that of $X_{9,t}$. In the conventions (5.41),*

$$G^{\text{res}}|_{\Sigma} = G^{\text{sm}}, \quad F^{\text{res}}|_{\Sigma} = F^{\text{sm}}, \quad F_{\text{inst}}^{\text{res}}|_{\Sigma} - F_{\text{inst}}^{\text{sm}} = -\frac{4\zeta(3)}{(2\pi i)^3}. \quad (5.45)$$

The integral map sends the surviving classes $(\mathbf{m}'_0, \mathbf{m}'_2, \mathbf{m}'_3, \mathbf{m}'_4; \mathbf{e}_0, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4)$ to the correspondingly ordered smooth-side basis, in $V^\perp/V$.

Proof. Conifold excision identifies H_3 of the complement of the vanishing spheres with $V^\perp$. Filling by the small-resolution neighborhoods quotients it by the images of the boundary three-spheres, namely V . Since V is primitive, this is the free integral symplectic quotient $V^\perp/V$. The residue-preserving birational identification in Proposition 5.2 preserves this pairing. The corrected basis (5.40) makes the quotient explicit: discard $\mathbf{m}'_1, \mathbf{m}'_5, \mathbf{m}'_6$ and quotient by $\mathbf{e}_1, \mathbf{e}_5, \mathbf{e}_6$. The eight classes stated in the proposition give a unimodular symplectic basis.

The D6 class $\mathbf{m}_0$ lies in $V^\perp$, since every vanishing charge is pure D2. Its period thus descends. The difference of its descended class and the smooth-side $\mathbf{m}_0$ pairs trivially with each of the seven already identified classes. Their common annihilator is the line spanned by $\mathbf{e}_0$. Consequently

$$W(G^{\text{res}}|_\Sigma - G^{\text{sm}}) = cW, \quad c \in \mathbb{Z}. \quad (5.46)$$

This step uses the polarization, not just the dimension of the period space. A real coefficient would already suffice below.

We determine c on the positive real branch. Subtract the quadratic term from the prepotential and write $\tilde{F}_a = F_a - a_{ab}t_b$; this rational symplectic change does not alter G . The electric and second-derivative Abel continuation in Appendix C, along $z_1 = z_5 = z_6 = \rho \uparrow 1$ with fixed sufficiently small positive (v, a, b) , has

$$W \in \mathbb{R} \setminus \{0\}, \quad t_a \in i\mathbb{R}, \quad \tilde{F}_a \in \mathbb{R}. \quad (5.47)$$

The Gamma coefficients and logarithms are real before their explicit powers of $2\pi i$ are inserted. The convergent weighted sums fix these reality properties on the continuation, rather than selecting a new branch at the endpoint. The initial large-volume value of G is purely imaginary by (5.41); the quadratic term cancels from $2F - t_a F_a$. Special geometry gives

$$dG = \sum_a (\tilde{F}_a dt_a - t_a d\tilde{F}_a). \quad (5.48)$$

Its right-hand side is purely imaginary along this path. Hence the continued G^{res} is purely imaginary, including its finite endpoint supplied by conifold descent. This argument requires no termwise limit of an uncontracted third-derivative series. The smooth-side G^{sm} is likewise purely imaginary on its positive small-parameter branch. Equation (5.46) then forces the real integer c to vanish. Analytic continuation from this open real locus gives the identity throughout the marked complex neighborhood.

The seven-period result, together with (5.43), makes all three derivatives of $F^{\text{res}}|_\Sigma - F^{\text{sm}}$ zero. Equality of G fixes this constant to zero. Finally $\chi(\hat{X}_9) = -232$ and $\chi(X_{9,t}) = -240$; subtracting the perturbative terms in (5.41) gives the last equality in (5.45). $\square$

Scalar and electromagnetic couplings. The eighth-period match determines the two-derivative vector action, including its coupling to the graviphoton. In the common normalization,

$$e^{-K_{\text{vec}}} = i(\overline{X^A} F_A - X^A \overline{F_A}), \quad (X^A) = (1, \tau_i), \quad (F_A) = (G, F_i), \quad (5.49)$$

has the same restriction on both sides. Consequently the scalar kinetic metric $g_{i\bar{j}} = \partial_{\tau_i} \partial_{\bar{\tau}_j} K_{\text{vec}}$ agrees. The terms involving the three vanishing electric coordinates drop out of the original pairing.

Set $\mathcal{F}(X) = (X^0)^2 F(X^i/X^0)$ for the homogeneous prepotential, and write $\mathcal{F}_{AB} = \partial_A \partial_B \mathcal{F}$ for $A, B = 0, \dots, 3$. The supergravity period matrix is

$$\mathcal{N}_{AB} = \overline{\mathcal{F}}_{AB} + 2i \frac{(\text{Im } \mathcal{F}_{AC}) X^C (\text{Im } \mathcal{F}_{BD}) X^D}{X^E (\text{Im } \mathcal{F}_{EF}) X^F}. \quad (5.50)$$

In the convention where $-\text{Im } \mathcal{N}$ is the positive gauge-kinetic form, $\text{Re } \mathcal{N}$ gives the theta couplings [33, Equations (1.1) and (4.22)]. Equality of the reduced prepotentials therefore identifies both matrices, including graviphoton mixing, in the stated integral electric frame. These are the surviving couplings after conifold reduction, not the divergent charged block in (5.11).

The eighth period matters because a constant in the affine F becomes a quadratic term in the homogeneous $\mathcal{F}$. Its imaginary part can change K_{vec} and $\mathcal{N}$, even though it leaves the affine Yukawa couplings unchanged. Real quadratic changes also affect the electromagnetic frame. The integral marking and constant matching remove these ambiguities. This determines the genus-zero vector-sector couplings on the marked branch, not a hypermultiplet metric or a higher-derivative effective action.

5.7 The auxiliary-circle limit

The marked four-node locus has a large-radius limit on the original five-dimensional contraction face. Its four-hypermultiplet infrared description, however, has no radius-independent range of validity. These are distinct statements: the limiting geometry can be identified without treating the finite-circle conifold theory as the full five-dimensional theory.

Suppression of smooth-side instantons. The smooth-side effective cone gives a further all-orders consequence. By (5.25), every nonzero effective degree has $m + n > 0$. Hence $J_0 = \alpha K_3 + \beta K_2$, with $\alpha, \beta > 0$, lies in the interior of the smooth-side Kähler cone: its degree on a curve is $\alpha n + \beta m$. The ambient boundary curves of degree $(1, 0, 0)$ do not meet the Calabi–Yau hypersurface. Along (5.53), every smooth-side instanton is thus suppressed, although $v = v_0$ remains fixed. The convergent Frobenius series and the locally invertible mirror map give an analytic nonpolynomial prepotential near the origin. To see the convergence directly, the factorial ratios and their first two derivatives are bounded by an exponential in total degree times a polynomial in that degree; the latter includes the harmonic-number factors. Analytic inversion of the mirror map and integration of the regular magnetic series preserve a nonzero convergence domain. Its support has no pure q_1 term, so, for any fixed number of flat-coordinate derivatives,

$$\partial_{\tau_{i_1}} \cdots \partial_{\tau_{i_k}} F_{\text{inst}}^{\text{sm}} = O\left(e^{-2\pi L \min(\alpha, \beta)}\right). \quad (5.51)$$

The bound holds at fixed sufficiently small v_0 on the smooth-side large-volume branch. Consequently its quantum vector couplings have the cubic five-dimensional limit already matched in (3.18).

Radius and period normalization. Let R_{aux} be the radius of the spacetime circle reducing five dimensions to four, measured in the five-dimensional metric, and put $\mathcal{R} = R_{\text{aux}} M$ for a fixed mass unit M . This is not the elliptic-fiber circle of Section 6. In the leading circle dictionary the dimensionless vector coordinates are $t_a = B_a + i\mathcal{R}h_a$, where the real five-dimensional scalars obey $\frac{1}{6}\kappa_{abc}h_a h_b h_c = 1$ after volume normalization [34, eqs. (5.35)–(5.41)]. The total M-theory Calabi–Yau volume is a separate hypermultiplet parameter and is kept fixed and finite here. Calibrating the periods by the unit D0 charge gives, for an existing BPS state,

$$M_\gamma^{(5\text{-metric})} = \frac{|Z_\gamma|}{R_{\text{aux}}}. \quad (5.52)$$

The four-dimensional Einstein-frame mass has an additional common Weyl factor; it is not the mass used in (5.52). The same KK normalization is used for local circle theories in [35, eq. (2.27)].

A scaling inside the marked family. Fix sufficiently small positive v_0, a_0, b_0 and constants $\alpha, \beta > 0$. With $L \rightarrow \infty$, take

$$z_1 = z_5 = z_6 = 1, \quad v = v_0, \quad a = a_0 e^{-2\pi L\beta}, \quad b = b_0 e^{-2\pi L\alpha}. \quad (5.53)$$

Choose generic constants so that the path lies in the four-node stratum for all sufficiently large L . This path stays in the convergence domain of the marked period resummation. Writing $t_j = iy_j$ for $j = 2, 3, 4$, its exact flat coordinates satisfy

$$\begin{aligned} t_1 = t_5 = t_6 = 0, \quad y_2 = c + O(a + b), \quad c = -\frac{\log v_0}{2\pi} > 0, \\ y_3 = L\beta + O(1), \quad y_4 = L\alpha + O(1). \end{aligned} \quad (5.54)$$

Indeed all logarithmic corrections are bounded analytic functions of (v, a, b) by (C.27). At $a = b = 0$, (C.10) gives $q_2 = v$, which fixes the first limit in (5.54). Thus, for $J_\Sigma = y_2 H_2 + y_3 H_3 + y_4 H_4$,

$$\frac{J_\Sigma}{L} \longrightarrow J_0 = \alpha D_0 + \beta D_1, \quad \frac{J_\Sigma^3}{6L^3} \longrightarrow \mathcal{V}_0 = \frac{83\alpha^3 + 147\alpha^2\beta + 81\alpha\beta^2 + 14\beta^3}{6} > 0. \quad (5.55)$$

The leading radius therefore scales as $\mathcal{R} \sim L\mathcal{V}_0^{1/3}$, and the volume-normalized five-dimensional class tends to $J_0/\mathcal{V}_0^{1/3}$. Bounded resummed magnetic corrections do not alter this leading cubic scaling. We do not assume that the local instanton corrections themselves vanish at the conifold locus.

The extra condition needed to contract the whole del Pezzo surface is visible in its restriction lattice. In the basis (h, e_1, e_2) of $E = \text{Bl}_2 \mathbb{P}^2$,

$$H_2|_E = h, \quad H_3|_E = H_4|_E = 0, \quad (D_8 \cap E) = \beta_2. \quad (5.56)$$

The two exceptional curves have zero period on Σ , whereas β_2 has period $iy_2 \neq 0$. Its rescaled area y_2/L tends to zero, so $J_\Sigma|_E/L \rightarrow 0$: the entire surface, not only the two exceptional curves, contracts. This is precisely the additional condition distinguishing the five-dimensional face (4.3) from the codimension-three complex four-node locus.

Finite bulk gaugings and the local strong-coupling limit. The limit in (5.55) does not send the M-theory bulk volume to infinity. In particular it is not the rigid limit of Section 4. With fixed five-dimensional Planck mass, dimensional reduction instead gives $M_4^2 = 2\pi R_{\text{aux}} M_5^3$ in the common five-dimensional metric. The two vectors imposing cooperation remain normalizable. This can be checked after removing the graviphoton from the Hodge metric:

$$\mathcal{G}_J^{\text{vec}}(P, Q) = -JPQ + \frac{(J^2 P)(J^2 Q)}{6\mathcal{V}_J}. \quad (5.57)$$

Up to the common finite volume normalization, the block on $(V_1, V_2) = (-D_6, D_5)$ at J_0 is

$$\mathcal{G}_{J_0}^{\text{vec}}|_{V_1, V_2} = \begin{pmatrix} d & \tau - d \\ \tau - d & d \end{pmatrix}, \quad d = \frac{2(7\alpha^2 + 8\alpha\beta + 2\beta^2)^2}{3\mathcal{V}_0}, \quad \tau = 3\alpha + 2\beta. \quad (5.58)$$

Its eigenvalues are τ and

$$2d - \tau = \frac{143\alpha^4 + 289\alpha^3\beta + 199\alpha^2\beta^2 + 52\alpha\beta^3 + 4\beta^4}{83\alpha^3 + 147\alpha^2\beta + 81\alpha\beta^2 + 14\beta^3}, \quad (5.59)$$

both finite and strictly positive. By contrast, put $\eta = y_2/L$ and $J = J_0 + \eta H_2$. The restriction data give

$$\frac{1}{2}J^2 E = \frac{\eta^2}{2}, \quad \mathcal{G}_J^{\text{vec}}(E, E) = 3\eta + \frac{\eta^4}{6\mathcal{V}_J} \longrightarrow 0. \quad (5.60)$$

Bounded H_3, H_4 offsets in (5.54) do not change these limits. Thus the del Pezzo string tension scale vanishes and its Coulomb direction becomes strongly coupled, while the two bulk gaugings remain dynamical. These are limiting five-dimensional geometric couplings, not the four-dimensional threshold couplings obtained by integrating out massless conifold hypermultiplets.

Why four hypermultiplets do not suffice in the limit. The KK gap is $1/R_{\text{aux}}$. At any fixed positive energy cutoff, the number of Fourier modes of even one five-dimensional massless field grows without bound. Hence the four-dimensional truncation requires energies much smaller than $1/R_{\text{aux}}$ and has no uniform cutoff as $R_{\text{aux}} \rightarrow \infty$.

Equations (5.53)–(5.55) identify the limiting Coulomb-branch locus with the original five-dimensional cooperative contraction. They do not construct the interacting five-dimensional Higgs action or its finite-Planck metric. The choice of mirror crepant chamber and magnetic BPS survival also remain separate questions.

6 Fibrations and the six-dimensional interpretation

6.1 The K3-fiber vector

The two vectors imposing the joint constraint have a fibration interpretation. The functional $-e_3^*$ on the fan takes value 1 on v_5 , -1 on v_3, v_7, v_8 and zero elsewhere; no resolved cone contains rays of both signs. It defines a K3 fibration

$$\rho : \widehat{X}_9 \longrightarrow \mathbb{P}^1, \quad [\text{K3}] = D_5 \sim D_3 + D_7 + D_8. \quad (6.1)$$

The generic fiber has Picard rank three. Its classes $(D_2|, D_6|, E|)$ have Gram matrix

$$\begin{pmatrix} -2 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix}, \quad \text{Pic}(\text{K3}) \simeq U \oplus \langle -4 \rangle. \quad (6.2)$$

The middle class is an elliptic fiber and the other two are disjoint sections. The horizontal surface D_6 is a K3 blown up once; its exceptional curve is C_1 . In contrast, C_2 is a section of ρ .

In the K3-fibration heterotic/type-II dictionary, D_5 is the heterotic dilaton vector multiplet [36–39]. Since $D_5^2 = 0$, its norm is

$$\mathcal{N}_{D_5} = \frac{\text{vol}(\text{K3})^2}{\mathcal{V}}. \quad (6.3)$$

Thus one of the shared gaugings is a dilaton gauging; the other, $-D_6$, restricts to the elliptic fiber class in the K3 lattice. Vertical versus horizontal M2 curves give the usual perturbative versus nonperturbative interpretation of C_1 and C_2 in this dictionary. This interprets the compact charges but does not independently construct their heterotic states or condensates.

6.2 The flat elliptic model and its anomalies

The nef divisor $w = D_4 + D_5 = D_3 + D_6$ has $w^3 = 0$ and defines a genus-one fibration over $\mathbb{P}^2$. Its fiber $F = w^2$ pairs with (D_0, D_1, D_2, E) as $(3, 2, 1, 1)$ and trivially with the other divisors. There are two rational sections D_2, E and two non-flat points q_3, q_4 on a line L_0 . Flopping C_1 gives a flat elliptic threefold X_9^{fl} over $B' = \text{Bl}_{q_3, q_4} \mathbb{P}^2$: the fan projection to the last two coordinates then maps every cone to a cone. Write H, E_3, E_4 for the base hyperplane and exceptional classes, and $L = H - E_3 - E_4$.

The fiber monomials give a Morrison–Park two-section model [40], specialized so that L carries an I_2 fiber and the sections are disjoint. The latter condition removes charge-two singlet loci in the normalization where the ordinary singlet has charge one. The flat six-dimensional tensor branch has $T = 2$, algebra $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$, and gauge curve L . The two exceptional base curves support the geometric E-string sectors [41]. Its vector count is $T + \text{rank } G + 1 = 5$, as required by (3.5).

The charged matter and anomaly coefficient are

$$10 \mathbf{2}_{1/2} + 76 \mathbf{1}_1, \quad b = \frac{11}{2}H - \frac{3}{2}E_3 - \frac{3}{2}E_4. \quad (6.4)$$

Here the gravitational hypermultiplet count weights a doublet by dimension two. Thus $H_{mtot} = 123 + 20 + 76 = 219$, $V = 4$, and $H_{mtot} - V + 29T = 273$. The nonabelian anomaly conditions use $K_{B'}L = L^2 = -1$ and give ten fundamentals. In the conventional Abelian normalization [42, 43], the remaining checks read

$$\begin{aligned} -6K_{B'}b &= 81 = 20(1/2)^2 + 76, \\ 3b^2 &= \frac{309}{4} = 20(1/2)^4 + 76, \\ bL &= \frac{5}{2} = 10(1/2)^2. \end{aligned} \quad (6.5)$$

These agree with the geometric count of 76 isolated I_2 fibers away from L_0 .

The full height, including its exceptional-base components, is independently fixed by the sections. Put $P = \pi^*c_1(B') = 3w - D_3 - D_4$ and use D_2 as zero section. The Shioda and shifted Kaluza–Klein divisors are

$$\sigma = E - D_2 - P + \frac{1}{2}D_8, \quad Z = D_2 + \frac{1}{2}P, \quad -\pi_*(\sigma^2) = b. \quad (6.6)$$

They obey $\pi_*(\sigma Z) = \pi_*(\sigma D_8) = \pi_*(Z^2) = 0$ [44]. The half-Cartan term supplies the geometric center identification: in integral charge normalization the doublets have charge one and the singlets charge two, giving $U(2) = (SU(2) \times U(1))/\mathbb{Z}_2$. These statements come from the F-theory geometry, not a heterotic bundle tuning.

6.3 The two E-string roots and the elliptic-circle limit

On X_9^{fl} the surfaces $S_i = D_i$, $i = 3, 4$, are rational elliptic surfaces. Let $o_i = D_2|_{S_i}$, $s_i = E|_{S_i}$, $F_i = -K_{S_i}$, and define

$$r_i = s_i - o_i - F_i. \quad (6.7)$$

The two sections are disjoint, have square -1 and meet the fiber once. Hence $r_i^2 = -2$ and $r_i o_i = r_i F_i = 0$: each is a root in the $E_8(-1)$ orthogonal complement of $\langle o_i, F_i \rangle$ [45]. Their compact images coincide. Indeed $o_3 = -C_1$, $o_4 = C_2$, $F_3 = F_4 = F$ and $s_4 - s_3 = R$, so

$$r_4 - r_3 = R - C_1 - C_2 = 0. \quad (6.8)$$

The integral curve pairings verifying these identities are printed in Appendix D. This identifies two local root images, not the entire microscopic heterotic E_8 gauge algebras or a BPS state for every signed root class.

The transition face also determines whether a six-dimensional limit exists. Write $J = aD_0 + bD_1$ with $a, b \geq 0$ and $a + b > 0$, and $\tau = JF = 3a + 2b$. The original and flat chambers share this face because $JC_1 = 0$. The cubic gives

$$\frac{7}{24} \leq \frac{\mathcal{V}}{\tau^3} \leq \frac{83}{162}. \quad (6.9)$$

To prove the bound, put $x = a/\tau \in [0, 1/3]$. Then

$$\frac{\mathcal{V}}{\tau^3} = \frac{7}{24} + \frac{3}{4}x - \frac{1}{8}x^2 - \frac{5}{12}x^3. \quad (6.10)$$

Its derivative is decreasing and has positive minimum $19/36$; the endpoint values give (6.9). The F-theory limit requires $\tau/\mathcal{V}^{1/3} \rightarrow 0$ [46], so it is excluded on this face. This is the elliptic 6d-to-5d circle, not the auxiliary 5d-to-4d circle of Section 5.7. Nor does the bound force the total Planck volume to vanish at fixed fiber volume: the section E has zero volume on the face, but $\text{vol}(D_2) = 2a^2$.

For completeness the finite-circle charge relation is visible without taking either limit. Expand $J = \tau Z + \pi^* j + z\sigma + \phi D_8$; on the face, $z = 2a + b$ and $\phi = -b/2$. The common root image has $(Zr, \sigma r, D_8 r) = (0, -3/2, 1)$, so

$$J(r + F) = \tau - \frac{3}{2}z + \phi = 0. \quad (6.11)$$

The section-difference class has one unit of this elliptic Kaluza–Klein charge. A different zero section changes the divisor coordinates, not the volume bound. At finite radius, j is not directly the six-dimensional base Kähler class: the tensor dictionary includes Green–Schwarz shifts [44, 46].

6.4 The status of the heterotic construction

Regard B' as $\mathbb{F}_1$ blown up once on the positive section. Its heterotic parent has instanton numbers (11, 13); the extra blowup replaces one instanton by a five-brane [47, 48]. The residual data are

$$(k_1, k_2; N_5) = (11, 12; 1), \quad k_1 + k_2 + N_5 = 24. \quad (6.12)$$

In this presentation the common root above identifies a root of the E-string over the negative section with one of the five-brane’s E-string. It does not derive their condensate dynamics.

An integral subdivision of the Newton polytope along $m_4 + 2m_1 + m_2 = 0$ has a reflexive common slice with 30 lattice points. It produces a candidate elliptic heterotic K3 with a distinguished I_2 fiber; an exact member has fibers $I_2 + 22I_1$. Its equation and coefficient data are given in Appendix D. The semistable total family, bundle or spectral-sheaf data and their gluing remain to be constructed. In particular the massless $U(1)$ requires a Stückelberg check, not merely a bundle commutant [49, 50]. The F-theory spectrum and root relation therefore constrain a heterotic construction; they do not complete it.

7 Discussion

The transition in X_9 exhibits a compact coupling between two conifold sectors and an interacting E_2 sector. Its charge relation constrains the invariant Higgs coordinates, its integral neutral lattice fixes the surviving vector fields, and its mirror periods determine their genus-zero quantum couplings. The four-hypermultiplet thresholds reproduce the singular

monodromy, while the eight-period comparison fixes the regular scalar metric and electromagnetic couplings of the neutral sector. Absence of neutral logarithmic thresholds alone would not establish that comparison.

The two limits explain the scope of this effective description. With bounded local Kähler scales, sending the compact volume to infinity removes the shared flavor gauging. Sending the auxiliary radius to infinity at fixed five-dimensional Planck scale retains finite bulk gauging but closes the Kaluza–Klein gap. Neither limit produces an autonomous four-particle description retaining the original cooperative constraint.

Three extensions would materially strengthen the physical description. First, an explicit finite-Planck hypermultiplet geometry would test how the rigid invariant-coordinate description is embedded in quaternionic-Kähler geometry. Second, a semistable heterotic degeneration with bundle data would determine the microscopic realization of the shared gauging, including the masslessness of its $U(1)$. The K3 slice and topological counts alone do not provide this. Third, a stability condition along the quantum continuation would distinguish which magnetic sheaves remain physical particles and which can decay. The isolated numerical Donaldson–Thomas contributions and partial wall exclusions in [16] do not determine that spectrum.

Geometric existence is governed by the exact mixed smoothing criterion, including the three discriminant conditions on the $SU(3)$ component of the E_3 branch [8]. Selected-base integrability under its projection hypothesis removes a geometric obstruction; it does not construct the compact hypermultiplet action. Questions about selected profiles with zero E_3 projection and unrestricted nonreduced deformation spaces remain outside that theorem. On the quantum side, continuation beyond the marked branch and higher-genus transition formulas are natural next problems. The results proved here require neither of these extensions: they establish one actual mixed compact transition, the obstruction to retaining its flavor gauging in the specified rigid limits, and the complete genus-zero vector geometry across its circle-reduced transition.

A Local-to-global deformation results

A.1 Local rings and branch markings

For the E_2 cone the full current ring, including the nilpotent operator U , is [11]

$$\frac{\mathbb{C}[J_+, J_0, J_-, U]}{(J_0^2 - J_+ J_-, U^2, U J_+, U J_0, U J_-)}. \quad (\text{A.1})$$

Every weight-zero monomial not containing U reduces to a power of J_0 ; every nonzero monomial containing U is proportional to U . The Cartan invariants are therefore

$$\frac{\mathbb{C}[J_0, U]}{(U^2, U J_0)} \simeq \frac{\mathbb{C}[\beta, \alpha]}{(\alpha^2, \alpha\beta)}. \quad (\text{A.2})$$

The right side is Altmann's miniversal deformation ring of the degree-seven cone [12]. Its root marking permits

$$\alpha = U, \quad \beta = J_0 + cU \quad (\text{A.3})$$

with compatible nonzero rescalings. The shear preserves the ring because $U^2 = UJ_0 = 0$. The nilpotent direction vanishes at every ordinary complex point and is not a second reduced Higgs branch.

For $S_6 = \text{Bl}_3 \mathbb{P}^2$, the reduced Higgs scheme is the union at the origin of the minimal nilpotent orbit closures of $\mathfrak{sl}_3$ and $\mathfrak{sl}_2$ [11]. Their torus-invariant rings are $\mathbb{C}[x_1, x_2]$ and $\mathbb{C}[l]$, giving

$$\mathbb{C}[\mathcal{H}_{E_3}]^{T_{\mathbb{C}}} = \mathbb{C}[x_1, x_2] \times_{\mathbb{C}} \mathbb{C}[l] = \frac{\mathbb{C}[x_1, x_2, l]}{(x_1 l, x_2 l)}. \quad (\text{A.4})$$

This equals the miniversal ring of the degree-six cone [12]. The two components select the sublattices

$$A_2 = \langle e_1 - e_2, e_2 - e_3 \rangle, \quad A_1 = \langle h - e_1 - e_2 - e_3 \rangle \subset K_{S_6}^{\perp}. \quad (\text{A.5})$$

To verify the invariant statement and balanced lifts, write a rank-one traceless matrix as $q_i \tilde{q}_j$ with $z_i = q_i \tilde{q}_i$ and $\sum_i z_i = 0$. Its torus invariants are generated by the diagonal entries: every closed monomial cycle reduces to their product by rank one. Choosing $|q_i| = |\tilde{q}_i| = \sqrt{|z_i|}$ and the appropriate phases sets the real ADHM and flavor Cartan moment maps to zero. For the A_2 root coordinates take $(z_1, z_2, z_3) = (x_1, x_2 - x_1, -x_2)$; the A_1 case uses two pairs. This supplies the local lifts used in Proposition 2.1.

A.2 Links and the obstruction map

Let Σ_p be the singularity link and F_{p, σ_p} a Milnor fiber on the chosen branch. Define

$$V_{p, \sigma_p} = \ker(H_2(\Sigma_p; \mathbb{Q}) \longrightarrow H_2(F_{p, \sigma_p}; \mathbb{Q})), \quad K_{\sigma} = \ker(\bigoplus_p V_{p, \sigma_p} \longrightarrow H_2(\widehat{X}; \mathbb{Q})). \quad (\text{A.6})$$

For a node the link space and V are both $\mathbb{Q}$. For a del Pezzo cone the Gysin sequence of the unit-circle bundle of K_{S_p} gives $H_2(\Sigma_p; \mathbb{Q}) = K_{S_p}^{\perp}$. The branch subspaces are the A_1 line for S_7 and the A_1 line or A_2 plane for S_6 . These are exactly the cycles killed in the corresponding Milnor fibers [19, Proposition 2.1 and Section 2.4]. Thus $K_{\sigma} \otimes \mathbb{C} = \ker Q_{X, \sigma}$.

We use the following local comparison in the proof of Theorem 2.2. For the chosen local crepant resolution with exceptional set E_p , let

$$K_p^{\text{cr}} = \ker(H_{E_p}^2(\widehat{X}_p; \Omega^2) \longrightarrow H^2(\widehat{X}_p; \Omega^2)). \quad (\text{A.7})$$

Contraction with the crepant volume form and the boundary map give compatible identifications

$$T_{X, p}^1 \simeq K_p^{\text{cr}} \simeq H_2(\Sigma_p; \mathbb{C}), \quad (\text{A.8})$$

carrying a branch tangent space to $V_{p, \sigma_p} \otimes \mathbb{C}$. The local-cohomology maps commute with pushforward of link classes. On the pentagon the smoothing line is the unique A_1 root line; on the hexagon the reflection and Coxeter eigenspaces give the A_1 and A_2 summands. The

local comparison and its volume-form determinant twist are proved in [8, Lemmas 2.3–2.4], using Schlessinger’s comparison in the depth-three form [51, 52].

Proof of Theorem 2.2. For these normal Gorenstein threefolds, the cotangent-complex and differential deformation groups agree through degree two. The non-lci locus is finite [8, Lemma 2.2]. The local-to-global sequence is

$$0 \longrightarrow H^1(X, \Theta_X) \longrightarrow \mathbb{T}_X^1 \xrightarrow{\text{loc}} \bigoplus_p T_{X,p}^1 \xrightarrow{\text{ob}} H^2(X, \Theta_X). \quad (\text{A.9})$$

Excision on $X_{\text{reg}} = \widehat{X} \setminus E$ gives

$$H^1(X_{\text{reg}}, \Omega^2) \longrightarrow \bigoplus_p H_{E_p}^2(\widehat{X}_p, \Omega^2) \longrightarrow H^2(\widehat{X}, \Omega^2). \quad (\text{A.10})$$

Under (A.8), the last map followed by de Rham comparison is the homology pushforward, followed by Poincaré duality. Its source is $\bigoplus_p H_2(\Sigma_p; \mathbb{C})$ and its target is $H_2(\widehat{X}; \mathbb{C})$. A class in $K_\sigma \otimes \mathbb{C}$ has zero de Rham image. The $\partial\bar{\partial}$ -lemma for the projective resolution makes $H^2(\widehat{X}, \Omega^2) \rightarrow H^4(\widehat{X}; \mathbb{C})$ injective, so it lifts through (A.10). Contraction with the trivial canonical form identifies the first group there with $H^1(X_{\text{reg}}, \Theta)$, and the depth-three comparison identifies this with $\mathbb{T}_X^1$. Naturality gives precisely the localization map in (A.9). This proves one inclusion. Conversely, a global deformation lies in $\ker(\text{ob})$; the same comparison sends its branch link class to zero in $H_2(\widehat{X}; \mathbb{C})$. This proves the other inclusion. The admissible-polytope hypotheses ensure that ambient divisors span the full rational divisor space, so their finite pairing matrix imposes every continuous constraint [19, Proposition 8.3]. $\square$

A.3 Exact smoothing, selected bases and the multiplet change

Theorem A.1 (mixed smoothing criterion). *Let X be connected, normal and projective with $\omega_X \cong \mathcal{O}_X$ and $h^1(X, \mathcal{O}_X) = 0$, with only nodes and exact analytic anticanonical S_7, S_6 cones. Choose a smooth crepant resolution, small at the nodes and standard divisorial at the cones, and a reduced local profile σ . Then X has a convergent one-parameter smoothing on σ if and only if K_σ contains a vector nonzero on every rank-one summand and with all three coefficients μ_i nonzero on each A_2 summand, written as $\sum_i \mu_i e_i$ with $\sum_i \mu_i = 0$. Necessity applies to arbitrary analytic smoothing arcs.*

This is [8, Theorem 4.7] in the degree-six/seven case. The three A_2 coefficients are the nodal smoothing coordinates on its simultaneous partial resolution. In the root coordinates used here they are $(H_1, H_2 - H_1, -H_2)$, so the plane discriminant is $H_1 H_2 (H_1 - H_2) = 0$. The nodal period argument on the partial model proves their separate nonvanishing at every order; the scalar-period argument of [19] handles the rank-one germs. A finite union of proper linear subspaces cannot cover K_σ , so checking the individual scalar functionals is sufficient to find one relation avoiding all discriminants.

For analytic local classifying maps, let $\text{Def}^\sigma(X)$ be the scheme-theoretic inverse image of the selected reduced local branches. The companion integrability theorem states that, if $K_\sigma \rightarrow V_{p,\sigma_p}$ is nonzero at every E_3 point, then

$$\text{Def}^\sigma(X) \text{ is smooth,} \quad \dim \text{Def}^\sigma(X) = h^1(X, \Theta_X) + \dim K_\sigma, \quad (\text{A.11})$$

and every tangent on it is realized by a convergent curve [8, Theorem 4.5]. No extra projection hypothesis is required at nodes or E_2 points. Lifting a relation outside the discriminants and integrating it proves sufficiency in the preceding theorem. This does not assert smoothness of unrestricted deformation schemes, integrability on all zero-projection E_3 profiles, or an identification of a nonlinear classifying map with a linear inclusion. The geometric theorem also allows degree-five and $\mathbb{P}^1 \times \mathbb{P}^1$ cones, but their physical operator dictionaries are not used in this paper.

For one E_2 point there is a direct topological count that avoids any assumption about the unknown hypermultiplet metric. Let n be the number of nodes, $k = \dim K_\sigma$ and $\rho = n + 1 - k$.

Proposition A.2. *If the exact criterion holds and X_t is a smoothing, then*

$$h^{1,1}(\widehat{X}) - h^{1,1}(X_t) = \rho + 1, \quad h^{2,1}(X_t) - h^{2,1}(\widehat{X}) = k. \quad (\text{A.12})$$

Proof. Use the common complement Y , exceptional neighborhoods N_p and Milnor fibers F_p . The local data are

	$b_2(N_p)$	$\chi(N_p)$	$b_2(\Sigma_p)$	$b_2(F_p)$	$\chi(F_p)$
node	1	2	1	0	0
$C(S_7)$	3	5	2	1	1.

(A.13)

For S_7 , the Milnor fiber is $\text{Bl}_{pt} \mathbb{P}^3 \setminus S_7$, and its link-map kernel is exactly the root line; see the integral restriction calculation in Appendix B. The link maps into N_p are injective and $H_1(\Sigma_p; \mathbb{Q}) = 0$, so $b_2(\widehat{X}) = b_2(Y) + 1$. Excision and Lefschetz duality give $H_3(\widehat{X}, Y; \mathbb{Q}) = \bigoplus_p H^3(N_p; \mathbb{Q}) = 0$, so $H_2(Y; \mathbb{Q})$ injects into $H_2(\widehat{X}; \mathbb{Q})$. The joint link map into $Y \sqcup \bigsqcup_p F_p$ has kernel K_σ , hence rank $n + 2 - k$. Therefore

$$b_2(X_t) = b_2(Y) + 1 - (n + 2 - k) = b_2(\widehat{X}) - \rho - 1. \quad (\text{A.14})$$

Euler additivity gives $\chi(X_t) - \chi(\widehat{X}) = -2n - 4$. Using $\chi = 2(h^{1,1} - h^{2,1})$ proves the second equality. $\square$

B Integral lattices and classical data

B.1 The resolved chamber and intersection form

Adjoin $v_9 = p = (-2, -1, 0, 0)$ and $v_{10} = (-1, 0, 0, 0)$ to (3.1). The lower-facet triangulation with heights

$$h_i = b_i + (i + 1)/1009, \quad (b_0, \dots, b_{10}) = (10, 11, 3, 9, 2, 7, 1, 4, 8, -5, -3) \quad (\text{B.1})$$

fixes a complete smooth fan with 29 maximal cones. The divisor D_{10} misses the hypersurface. In the basis (3.6), with $D_9 = E$, the nonzero cubic entries are as follows; all permutations agree and all omitted entries vanish:

$$\begin{array}{c|c|c|c|c|c}
IJK & \kappa_{IJK} & IJK & \kappa_{IJK} & IJK & \kappa_{IJK} \\
\hline
222 & 8 & 266 & 1 & 667 & -1 \\
225 & -2 & 267 & 1 & 677 & -1 \\
226 & -3 & 569 & 1 & 699 & -2 \\
227 & -2 & 599 & -2 & 777 & -3 \\
256 & 1 & 666 & -1 & 778 & 4 \\
\hline
& & & & 788 & -6 \\
& & & & 888 & 8 \\
& & & & 889 & -1 \\
& & & & 899 & -1 \\
& & & & 999 & 7
\end{array} \tag{B.2}$$

The second-Chern row is $(-4, 24, 26, 18, -4, -2)$, and the remaining divisor classes are

$$\begin{aligned}
D_0 &= D_2 + 4D_5 + 4D_6 - 3D_7 - 2D_8 + 2E, \\
D_1 &= D_2 + 2D_5 + 2D_6 - D_7 - D_8 + E, \\
D_3 &= D_5 - D_7 - D_8, \quad D_4 = D_6 - D_7 - D_8.
\end{aligned} \tag{B.3}$$

These data determine every norm, charge pairing and cubic used in Sections 3–4.

B.2 Resolved and smoothed lattices

Order the resolved divisors as $(D_2, D_5, D_6, D_7, D_8, E)$, with $D_9 = E$. The six integral curve classes

$$(D_6 D_7, D_2 D_4, D_4 D_5, D_8 E, D_3 D_7, D_0 D_5)|_{\widehat{X}_9} \tag{B.4}$$

give the divisor–curve pairing, with curves as rows,

$$P = \begin{pmatrix} 1 & 0 & -1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 2 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}, \quad \det P = 1. \tag{B.5}$$

Since the six divisors are already a rational basis, any integral cohomology class has rational coordinates whose pairing with these curves is integral. Multiplication by P^{-1} shows that those coordinates are integral. Thus the displayed divisors generate $H^2(\widehat{X}_9; \mathbb{Z})/\text{tors}$, and the displayed curve classes generate its dual. This argument does not infer integral saturation from the rational Hodge-number count.

The Mayer–Vietoris description of the smooth fiber also holds on free integral lattices. The links have $H^1(L; \mathbb{Z}) = 0$. On the E_2 surface use (h, e_1, e_2) with intersection form $\text{diag}(1, -1, -1)$ and $K = -3h + e_1 + e_2$. For $V_7 = \text{Bl}_{pt} \mathbb{P}^3$,

$$H|_E = 2h - e_1 - e_2, \quad Q|_E = h - e_1 - e_2, \quad K = -2H|_E + Q|_E. \tag{B.6}$$

The two restrictions form an integral basis of $R^\perp$, where $R = e_1 - e_2$: their matrix on $(h, e_1 + e_2)$ has determinant -1 . Both K and the deleted divisor class $2H - Q$ are primitive. Therefore $H^2(V_7 \setminus E; \mathbb{Z}) \cong \mathbb{Z}^2/\mathbb{Z}(2, -1)$ maps isomorphically to $R^\perp/\mathbb{Z}K$ in the link cohomology, with no finite index. For the resolution, excision and Poincaré–Lefschetz duality give $H^3(\hat{X}_9, Y; \mathbb{Z}) \cong H_3(N; \mathbb{Z}) = 0$, since the exceptional neighborhoods retract onto E and the two nodal curves. Thus restriction to the common complement is surjective, with kernel $\mathbb{Z}E$; the nodal Milnor fibers have $H^2 = 0$. The class E is primitive by (B.5). The neutral lattice has integral basis

$$(A, B, D_8, E) = (D_2 + D_7, D_5 + D_6 - D_7, D_8, E). \quad (\text{B.7})$$

Indeed adjoining $-D_6, D_5$ gives a unimodular six-divisor basis. Quotienting by E gives (3.12).

B.3 The local correction to the cubic

Set $V_7 = \text{Bl}_{pt} \mathbb{P}^3$, with pullback hyperplane H and exceptional plane Q . The compactifying surface is $S_7 \in |2H - Q|$ and its complement is the Milnor fiber. The restrictions in (B.6) identify the surviving link classes with $R^\perp/\mathbb{Z}K_E$. Every inherited divisor in $\text{span}(D_0, D_1)$ restricts to zero on the exceptional surface and the links. It therefore has a compactly supported representative on Y ; all products with an inherited factor agree on both sides, as do its second-Chern pairings.

For the remaining correction, glue the Milnor fiber to the orientation-reversed exceptional neighborhood. As a cut and paste, the resulting closed six-manifold is $V_7 - \mathbb{P}(\mathcal{O} \oplus K_{S_7}^{-1})$. If $\lambda = D_8|_E = e$ is the exceptional curve on $S_7 = \text{Bl}_1(\mathbb{P}^1 \times \mathbb{P}^1)$, its lifts are Q on V_7 and $\pi^*\lambda$ on the projective bundle. They give

$$Q^3 = 1, \quad (\pi^*\lambda)^3 = 0, \quad p_1(V_7)Q = 4, \quad p_1(\mathbb{P}(\mathcal{O} \oplus K^{-1}))\pi^*\lambda = 0. \quad (\text{B.8})$$

Here $c_1(V_7) = 4H - 2Q$, $c_2(V_7)H = 6$ and $c_2(V_7)Q = 0$. Since $p_1 = -2c_2$ on a Calabi–Yau threefold, these equalities give $\Delta(\nu^3) = 1$ and $\Delta(c_2\nu) = -2$. The independence of the chosen resolved representative is explicit:

$$(D_8 + tE)^3 + (-2tH + (1+t)Q)^3 = 9. \quad (\text{B.9})$$

The second term is the local correction associated with the same boundary class, not an extra bulk divisor.

Geometrically, the (-1) -curve $D_8 \cap E$ contracts and the surface $D_8 \cong \mathbb{F}_1$ maps to a plane through the singularity. Its local cone over e is replaced in the smoothing by $Q \setminus e \cong \mathbb{C}^2$. Gluing along their common three-sphere gives a smooth plane of class ν . Adjunction then yields $\nu^3 = K_{\mathbb{P}^2}^2 = 9$ and $c_2(X_{9,t})\nu = c_2(\mathbb{P}^2) - 9 = -6$ independently.

B.4 The integral flavor action

The effective action on the reduced Higgs cone need not be faithful on all states. In particular the E_2 non-Abelian flavor factor is $SU(2)$, not $SO(3)$ [53, Section 3.9.1]. We check the

compact action on the geometric M2 charge lattice, using the structure-group criterion of [53, Section 2.1]: divide the product of gauge and flavor factors by the central subgroup acting trivially on those charges.

For $\gamma = dh + m_1 e_1 + m_2 e_2$, choose integral charges

$$q_g = -K \cdot \gamma = 3d + m_1 + m_2, \quad w = -R \cdot \gamma = m_1 - m_2, \quad q_a = h \cdot \gamma = d. \quad (\text{B.10})$$

Here q_g is the local Coulomb charge, w is the standard $SU(2)$ weight with root weight two, and q_a is a complementary Abelian charge convention, allowing shifts by the Coulomb generator. Their image is the index-two lattice $q_g + w + q_a \in 2\mathbb{Z}$. Hence the faithful structure group on this charge lattice is

$$\frac{U(1)_g \times SU(2)_F \times U(1)_a}{\langle (-1, -\mathbf{1}, -1) \rangle}. \quad (\text{B.11})$$

This specifies the geometric charge action, not additional ultraviolet flavor data or an independent identification of the six-dimensional $U(2)$ gauge factor.

The restrictions of the six compact divisors to E , as columns in (h, e_1, e_2) , are

$$T = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & -3 \\ 0 & -1 & 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 0 & -1 & 1 \end{pmatrix}. \quad (\text{B.12})$$

For $D|_E = xh + ye_1 + ze_2$, its action on a curve is

$$D \cdot \gamma = -\frac{y+z}{2} q_g + \frac{z-y}{2} w + \left(x + \frac{3}{2}(y+z) \right) q_a. \quad (\text{B.13})$$

In particular, the coefficient triples for the two effective vectors are

$$V_1 : \left(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right), \quad V_2 : \left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right). \quad (\text{B.14})$$

Each 2π loop closes by the central identification in (B.11). On the reduced current ring the Coulomb and complementary Abelian actions are trivial, leaving unit charges for J_+ under both vectors, as required in (B.16). Thus the reduced action is consistent without requiring a half-Cartan to be a cocharacter of $SU(2)$ alone.

Finally this marking makes explicit the nilpotent-coordinate freedom in (A.3). Let U be the current normalized for q_a . On the six compact divisors the coefficient of that current is $(0, -\frac{1}{2}, -\frac{1}{2}, 0, -2, 0)$, whereas the geometric α column is $(0, -2, 1, 0, -2, 0)$. Their difference is $\frac{3}{2}$ times the root column. The compatible invariant coordinates are

$$\alpha_{\text{geom}} = U, \quad \beta_{\text{geom}} = J_0 + \frac{3}{2}U. \quad (\text{B.15})$$

The relations $U^2 = UJ_0 = 0$ show directly that this shear preserves the full invariant scheme. On its reduction $U = 0$ it changes neither the cooperative kernel nor the dressed-instanton ring.

B.5 The formal invariant ring

In the primitive effective basis $(-D_6, D_5)$, the charges are

$$\begin{array}{c|cccccc} & h_1 & \tilde{h}_1 & h_2 & \tilde{h}_2 & J_+ & J_- & \beta \\ \hline V_1 & 1 & -1 & 0 & 0 & 1 & -1 & 0 \\ V_2 & 0 & 0 & 1 & -1 & 1 & -1 & 0. \end{array} \quad (\text{B.16})$$

Remove the neutral pairs $h_i \tilde{h}_i$ and $J_+ J_-$ from any invariant monomial. Charge neutrality leaves a power of $h_1 h_2 J_-$ or $\tilde{h}_1 \tilde{h}_2 J_+$. Together with $u_i = -\beta$ and $J_+ J_- = \beta^2$, this proves the presentation (3.10). Grading the hypermultiplet scalars by degree one and the currents by degree two gives the refined Hilbert series

$$H(t, z) = \frac{1 - t^8}{(1 - t^2)(1 - zt^4)(1 - z^{-1}t^4)} = \frac{1}{1 - t^2} \sum_{n \in \mathbb{Z}} z^n t^{4|n|}, \quad (\text{B.17})$$

where Y has instanton fugacity z . An instanton current of charge n must be dressed by $|n|$ fields from each node. Removing the local instanton currents forces $\beta^2 = 0$, so the reduced formal quotient becomes a point. This detects the local strong-coupling sector in the operator model; it is not a heterotic perturbative calculation or a compact BPS index.

B.6 Topological D-brane generators

For either threefold X , let L_a denote its ordered primitive divisor basis: $H_1, \dots, H_6$ on the resolution and K_1, K_2, K_3 on the smooth side. Put $c_a = c_2(X) \cdot L_a$ and $\kappa_{abc} = L_a L_b L_c$. Work in untwisted topological $K^0(X)$, with Dirac pairing minus the Euler pairing. Choose generators

$$\mathbf{m}_0 = [\mathcal{O}_X], \quad \mathbf{m}_a = [\mathcal{O}_X(L_a)] - [\mathcal{O}_X], \quad \mathbf{e}_0 = -[\mathcal{O}_{\text{pt}}], \quad \text{ch}(\mathbf{e}_a) = (0, 0, \beta_a, 0), \quad (\text{B.18})$$

where $L_a \cdot \beta_b = \delta_{ab}$. Sans-serif symbols denote charge generators, distinct from the exceptional curves e_1, e_2 . The sign of $\mathbf{e}_0$ gives normalized period $+1$, consistently with the earlier central-charge convention. In particular $\mathbf{m}_0$ has Mukai charge $(1, 0, c_2(X)/24, 0)$, not the unshifted cohomological unit.

These generators form the full integral charge lattice. On the smooth side, the vertex matrices of the seven-ray fan polytope and its polar both have nonzero Smith entries $(1, 1, 1, 1)$. The same property was established for the resolution. The fundamental-group and Brauer-group formulas therefore give torsion-free integral cohomology on both sides [54, Corollaries 1.9 and 3.9]. The untwisted Atiyah–Hirzebruch spectral sequence degenerates: its differentials are rationally zero and its groups are torsion-free. Its free associated graded groups are H^0, H^2, H^4, H^6 [55, Section 3.1]. Thus every primitive H^4 class has a rank-zero, first-Chern-zero K -theory lift. Its index is $\int_X \text{ch}_3 \in \mathbb{Z}$, so a point class removes its third Chern character. This constructs the $\mathbf{e}_a$ in (B.18); the remaining generators lift the primitive H^0, H^2, H^6 bases. No BPS representative of every generator is assumed. In particular the smooth-side products of two divisors generate an index-ten sublattice of H^4 , not all of H^4 .

B.7 The current-exchange matrices

For $J = J_* + \Lambda w$ with $J_* = (6, 18, 16, -11, -8, 7)$ in $\mathcal{D}$, both nodal areas are one. The cubic tensor above and (4.6) give

$$\begin{aligned} H_{D_0} &= \frac{1}{66\Lambda} \begin{pmatrix} 31 & -9 \\ -9 & 31 \end{pmatrix} + O(\Lambda^{-2}), \\ H_{D_1} &= \frac{1}{14} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \frac{1}{196\Lambda} \begin{pmatrix} 96 & -5 \\ -5 & 110 \end{pmatrix} + O(\Lambda^{-2}), \\ H_{D_0+D_1} &= \frac{1}{590\Lambda} \begin{pmatrix} 209 & -91 \\ -91 & 209 \end{pmatrix} + O(\Lambda^{-2}). \end{aligned} \tag{B.19}$$

Since $Gt = v/2$ and $t^\top Gt = 3\mathcal{V}$, graviphoton subtraction removes the matrix

$$(3\mathcal{V})^{-1} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = O(\Lambda^{-3}).$$

C Analytic continuation of mirror periods

C.1 Toric coordinates and Frobenius series

The six generators of the ambient nef cone restricted to $\widehat{X}_9$ form an integral basis. In the divisor basis $(D_2, D_5, D_6, D_7, D_8, E)$, write $(H_1, \dots, H_6)$ as the columns of

$$N = \begin{pmatrix} 0 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 5 & 5 \\ 0 & 1 & 2 & 4 & 4 & 5 \\ 0 & -1 & -1 & -3 & -3 & -3 \\ 0 & -1 & -1 & -2 & -2 & -2 \\ 0 & 0 & 1 & 2 & 2 & 2 \end{pmatrix}, \quad \det N = -1. \tag{C.1}$$

In particular $H_1 = D_5$, $H_2 = D_5 + D_4$, $H_3 = D_1$, $H_4 = D_0$, and $J_* = \sum_a H_a$. The nonnegative orthant in the dual coordinates is an outer Mori cone sufficient for the period expansion; its lattice generators need not all be effective on the hypersurface.

The facet-interior ray D_f misses the hypersurface. Its mirror monomial coefficient can be removed by an ambient automorphism [15, Section 3.1]. Removing this ray and using the regular triangulation induced by the same heights gives a complete simplicial fan with 24 maximal cones. Its nef cone in the six-dimensional divisor space is exactly the cone generated by (C.1); thus removing D_f has not changed the chosen chamber on the hypersurface. The

GLSM charge matrix, with columns ordered as $(D_0, D_1, D_2, D_3, D_4, D_5, D_6, D_7, D_8, E)$, is

$$Q = \begin{pmatrix} 0 & 0 & -1 & 0 & -1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & 1 \\ 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 1 & -1 \end{pmatrix}. \quad (\text{C.2})$$

Its rows are integral relations among the ten ray vectors; the augmented relation also has central-monomial coefficient $-Q_{a*}$, where $Q_{a*} = \sum_i Q_{ai} = (0, 0, 2, 1, 0, 0)_a$. For the mirror Laurent polynomial $f = a_* - \sum_i a_i x^{v_i}$, choose algebraic coordinates $z_a = \prod_i (a_i/a_*)^{Q_{ai}}$.

For $\mathbf{n} \in \mathbb{Z}_{\geq 0}^6$, put $k_i = \sum_a n_a Q_{ai}$ and $k_* = \sum_i k_i$. The Frobenius series is built from

$$c(\mathbf{n} + \boldsymbol{\rho}) = \frac{\Gamma(1 + \sum_a (n_a + \rho_a) Q_{a*})}{\prod_i \Gamma(1 + \sum_a (n_a + \rho_a) Q_{ai})}. \quad (\text{C.3})$$

Let $W_0 = \sum_{\mathbf{n}} c(\mathbf{n}) z^{\mathbf{n}}$, $C_a = \sum_{\mathbf{n}} \partial_{\rho_a} c(\mathbf{n} + \boldsymbol{\rho})|_0 z^{\mathbf{n}}$, and let $\hat{C}_{ab}$ use the second derivative with its universal $\zeta(2)$ multiple of $c(\mathbf{n})$ subtracted. Equivalently, all three rational coefficients are obtained by differentiating $c(\mathbf{n} + \boldsymbol{\rho})/c(\boldsymbol{\rho})$ at zero. For example, when all $k_i \geq 0$ this rational function is

$$\frac{\prod_{j=1}^{k_*} (j + \sum_a \rho_a Q_{a*})}{\prod_i \prod_{j=1}^{k_i} (j + \sum_a \rho_a Q_{ai})}. \quad (\text{C.4})$$

For a negative $k_i = -m$, replace that denominator product by the numerator product $\prod_{j=0}^{m-1} (\sum_a \rho_a Q_{ai} - j)$. Such terms may have zero scalar coefficient but nonzero first or second derivatives; discarding them would lose the conifold invariants. The mirror map and magnetic generating functions are

$$2\pi i t_a = \log z_a + C_a/W_0, \quad \sum_{\mathbf{n} > 0} n_a n_{\mathbf{n}}^0 \text{Li}_2(q^{\mathbf{n}}) = \frac{1}{2} \sum_{b,c} \kappa_{abc}^H \left(\frac{\hat{C}_{bc}}{W_0} - \frac{C_b C_c}{W_0^2} \right), \quad (\text{C.5})$$

where $\kappa_{abc}^H = H_a H_b H_c$ is obtained from (B.2) by the basis change (C.1). This is the standard Frobenius period construction and multicover extraction [15, Section 4.4 and Appendix A].

C.2 The local exceptional-curve marking

Relative to the interior point p , the five vertices of the pentagon have coordinates

$$(1, 0), (0, 1), (-1, 0), (0, -1), (1, 1)$$

in the saturated lattice generated by $v_3 - p, v_4 - p$. The face polynomial is therefore $a_9 + a_3 \xi + a_4 \eta + a_5/\xi + a_6/\eta + a_8 \xi \eta$. With $u = z_1 z_5$, $v = z_2$, $w = z_6$, torus rescaling gives

$$1 + vx + y + uw/x + vw/y + xy = 0, \quad Y^2 = (vx^2 + x + uw)^2 - 4vwx^2(1 + x). \quad (\text{C.6})$$

The second equation follows by clearing xy in the first and taking the discriminant of the resulting quadratic in y . The small-parameter logarithmic correction is the function H below: $q_6 = we^{-H}$ and $q_2 = ve^{-H}$. Its terminating coefficient identity fixes the local endpoint and the exceptional-curve marking.

Exact endpoint and the two vanishing charges. Keep the local mass parameters variable and denote the logarithmic correction by

$$H(u, v, w) = \sum_{\substack{0 \leq m \leq \ell \leq k \leq \ell+m \\ \ell+k > 0}} \frac{(-1)^{\ell+k-1}(\ell+k-1)! u^m v^\ell w^k}{m!(\ell+m-k)!(\ell-m)!(k-m)!(k-\ell)!}. \quad (\text{C.7})$$

Thus $q_6 = we^{-H}$ and $q_2 = ve^{-H}$. At $u = 1$, summing over m by Vandermonde's identity gives

$$[v^\ell]H(1, v, w) = -\frac{(2\ell-1)!(2\ell)!}{(\ell!)^4} w^\ell {}_2F_1(-\ell, 2\ell; \ell+1; w), \quad \ell \geq 1. \quad (\text{C.8})$$

The terminating Chu–Vandermonde identity [56, eq. 15.4.24] evaluates the last factor at $w = 1$ as $(1-\ell)_\ell/(\ell+1)_\ell = 0$. This coefficient cancellation determines an analytic germ, not merely a formal substitution: for $|w| \leq 1$, the absolute coefficient at each v^ℓ is bounded by $(\ell+1)^2 125^\ell$, by the multinomial bound on each summand. Hence $H(1, v, 1) = 0$ for sufficiently small $|v|$.

For the real continuation to $v = w = 1$, the discriminant on $u = 1$ is

$$\text{disc}_x \mathcal{Q} = -256v^3w^4(w-1)^2\mathcal{P}(v, w), \quad \mathcal{P} = 16v^2w + (27w^2 - 18w - 1)v + 1. \quad (\text{C.9})$$

It has no additional zero in $0 < v \leq w < 1$. Indeed, the discriminant of $\mathcal{P}$ as a quadratic in v is $(w-1)(9w-1)^3$, which is negative for $1/9 < w < 1$. For $0 < w \leq 1/9$, $\mathcal{P}$ decreases on $0 \leq v \leq w$ and $\mathcal{P}(w, w) \geq 1 - w - 18w^2 \geq 2/3$. At $w = 1$, $\mathcal{P}(v, 1) = (4v+1)^2$. Thus continuation along this triangle connects the small- v germ to the diagonal real branch, without crossing another singularity. It follows that

$$H(1, v, 1) = 0 \quad (0 < v \leq 1), \quad \log q_6(1, 1, 1) = 0. \quad (\text{C.10})$$

The fiber itself factors at $u = w = 1$:

$$x(1+x)y^2 + (vx^2 + x + 1)y + vx = (xy + 1)(vx + xy + y). \quad (\text{C.11})$$

The two rational components meet transversely at $vx^2 - x - 1 = 0$, $y = -1/x$ for positive v . The two neck cycles have the same primitive class, up to orientation, in the compact elliptic curve: either component with small disks around its two nodes removed supplies a cylinder between them. Thus the electromagnetic monodromy is conjugate to T^2 . This statement forgets the flavor charge carried by the punctures.

The integral marking can be fixed before the collision. As $v \rightarrow 0$,

$$\frac{\text{disc}_x \mathcal{Q}}{-256u^2v^3w^4} = (w-1)(uw-1) + O(v), \quad q_6 = w + O(v). \quad (\text{C.12})$$

These are the exceptional-curve flop loci for e_2 and e_1 , respectively, since $q^{\iota_* e_2} = q_6$ and $q^{\iota_* e_1} = uq_6$. Following these two conifold branches fixes their primitive charges as $\gamma_i = (0, 0, \iota_* e_i, 0)$, in the large-volume marking. The reindexing $m \mapsto k - m$ in (C.7) gives the exact Weyl reflection

$$H(u, v, w) = H(u^{-1}, v, uw), \quad (\log u, \log q_6) \mapsto (-\log u, \log q_6 + \log u), \quad (\text{C.13})$$

on continuously chosen logarithmic branches. It exchanges the two exceptional-curve periods, with no D0 shift in this marking. For $u = 1 + d$, $v = w$ near 1, the two singularities split as

$$w_{e_1} = 1 - \frac{1 + \sqrt{5}}{2}d + O(d^2), \quad w_{e_2} = 1 + \frac{\sqrt{5} - 1}{2}d + O(d^2). \quad (\text{C.14})$$

C.3 The compact four-node family

For the finite-bulk calculation, use the monomial basis

$$m_x = (2, 1, 1, 0), \quad m_y = (2, 1, 0, 1), \quad m_r = (1, 0, 0, 0), \quad m_s = (1, 1, 0, 0). \quad (\text{C.15})$$

Its determinant is one. The last two rows of its inverse are $n_r = (1, -1, -1, -1)$ and $n_s = (0, 1, -1, -1)$, the inward normals to the two facets meeting in the pentagon. Their cone is smooth, and the segment joining them has no interior lattice points. Thus $(\mathbb{C}^*)^2 \times \mathbb{C}^2$, with regular coordinates r, s , is a valid chart of a crepant toric compactification. In particular, the loci $r = 0$ or $s = 0$ must not be discarded when looking for threefold singularities.

Write $w = z_6$, $v = z_2$, $a = z_3$, $b = z_4$. A coefficient gauge for the full Laurent polynomial is

$$a_* = 1, \quad (a_0, \dots, a_9) = (b, aw^2z_5^2, 1, v, 1, z_1z_5w, vw, (wz_5)^{-1}, 1, 1). \quad (\text{C.16})$$

Substitution in (C.2) gives exactly (z_1, v, a, b, z_5, w) . Multiplying the Laurent equation by its nonvanishing torus monomial and extending to this chart gives

$$\begin{aligned} \mathcal{F} = 1 + vx + y + \frac{z_1z_5w}{x} + \frac{vw}{y} + xy + r \left(1 + \frac{xy}{wz_5} \right) - rs \\ + br^2s + aw^2z_5^2s^2. \end{aligned} \quad (\text{C.17})$$

At $z_1 = z_5 = w = 1$, this is (5.3). Put $f = 1 + xy$, $g = 1 + x^{-1} + vy^{-1} + r$ and $h = r(br - 1) + as$. The four proposed singularities satisfy $f = g = s = h = 0$. At either root of $r(br - 1)$, $h_r = 2br - 1 = \pm 1$, while

$$\det \frac{\partial(f, g)}{\partial(x, y)} = x^{-1} + vx, \quad (x^{-1} + vx)^2 = (1 + r)^2 + 4v. \quad (\text{C.18})$$

Consequently (f, g, h, s) are analytic coordinates at each point and the equation is $fg + sh = 0$. Equivalently, the Hessian determinant in (x, y, r, s) is $(1 + r)^2 + 4v$. This proves the ordinary double point assertion to all orders in the bulk parameters.

The square charts have coordinates

$$(\xi_1, \eta, \alpha, \sigma) = ((ry)^{-1}, xy, r^{-1}, rs), \quad (\xi_2, \eta, \alpha, \sigma) = ((rx)^{-1}, xy, r^{-1}, rs). \quad (\text{C.19})$$

Their monomial bases, based at v_2 , are $(v_6 - v_2, v_7 - v_2, (-1, 0, 0, 0), (2, 1, 0, 0))$ and the same list with v_6 replaced by v_5 . Both are unimodular; all transverse monomial exponents are nonnegative. For positive b, v , the $r = b^{-1}$ roots satisfy $x_+ \sim (bv)^{-1}$ and $x_- \sim -b$ as $b \rightarrow 0$. Their respective limits in these charts are $(\xi_1, \eta, \alpha, \sigma) = (-v^{-1}, -1, 0, 0)$ and $(\xi_2, \eta, \alpha, \sigma) = (-1, -1, 0, 0)$, the two square critical points. The transverse derivative $\partial_\sigma \mathcal{F}_{\text{square}}$ at $\alpha = \sigma = 0$ is b . Thus the undeformed square-face critical points themselves are smooth points of the full threefold at $b \neq 0$; the actual nodes have moved into the adjacent facet orbit.

There is a nonempty open subset of this family with no further singularities. For example, take

$$(v, a, b) = (1/2, 1/100, 1/10). \quad (\text{C.20})$$

In the main chart the critical ideal, saturated by xy , is

$$(s, xy + 1, r(r - 10), \tfrac{1}{2}x^2 - (1 + r)x - 1) \subset \mathbb{Q}[x^{\pm 1}, y^{\pm 1}, r, s]. \quad (\text{C.21})$$

It is reduced of length four. Of the twelve facets of the Newton polytope, eleven have no torus-critical point at these coefficients. The remaining facet, with indices $(2, 3, 4, 5, 6, 7, 8, 9)$, is $s = 0$ and has precisely the critical curve $f = g = 0$ already treated. The only nonsimplicial two-faces are the pentagon and two squares; the latter are smooth in the full threefold by the derivative b . Every remaining lower-dimensional face has affinely independent support and nonzero coefficients. These exact algebraic checks exclude other singularities on a crepant toric resolution at (C.20). Properness and the ordinary double point condition then give a Zariski-open four-node family.

The relation and the deformation rank. The closure $W = \overline{\{s = f = 0\}}$ is an irreducible compact Weil divisor of the singular mirror. Near each node it is the plane $s = f = 0$ in $fg + sh = 0$, so its local divisor class is a generator, not a multiple. On W , the curve $g = 0$ is rational:

$$y = -x^{-1}, \quad r = -1 - x^{-1} + vx, \quad x^2 h = (vx^2 - x - 1)(b(vx^2 - x - 1) - x). \quad (\text{C.22})$$

The last polynomial has degree four, constant term b and leading coefficient bv^2 . Its four simple zeros are all the nodes. The two missing endpoints of the rational curve are covered by (C.19), where the normal derivative is nonzero. Thus W meets no additional singularity for generic bulk parameters. Cutting out four local balls and transporting the complement to a nearby smooth mirror produces a four-chain whose boundary is the sum of the four oriented vanishing spheres. This is the usual conifold four-chain construction [2, Section 3]; it gives a relation with four unit coefficients in absolute value. Blowing up W gives a projective small resolution of all four nodes and is an isomorphism elsewhere.

At a node, parameter differentiation gives (5.5); derivatives in v, a, b vanish. For the two roots at $r = 0$, write their inverse x -coordinates as $\rho \neq \rho'$. Either root at $r = b^{-1}$, with inverse coordinate τ , gives the minor

$$\det \begin{pmatrix} \rho & \rho & -1 \\ \rho' & \rho' & -1 \\ \tau & \tau + b^{-1} & -1 \end{pmatrix} = \frac{\rho - \rho'}{b} \neq 0. \quad (\text{C.23})$$

The leading period of a vanishing sphere is a nonzero multiple of its smoothing parameter. Thus these derivatives imply at least three independent homology classes. The four-chain relation gives at most three, proving rank three and uniqueness of the primitive relation.

One can also check the relative coefficients from residues. Up to a common sign, the holomorphic form in this chart is $\text{Res}(dx dy dr ds / (xy\mathcal{F}))$. Set $D = x^{-1} + vx$ at a node, so $x^{-1} = (D - 1 - r)/2$ and $D = \pm\sqrt{(1+r)^2 + 4v}$. After removing a common period normalization, its smoothing covector is

$$\frac{(x^{-1}, x^{-1} + r, -1)}{D(2br - 1)}. \quad (\text{C.24})$$

At fixed r , the sum over the two signs of D is $(1, 1, 0)/(2br - 1)$. The sums at $r = 0$ and $r = b^{-1}$ therefore cancel, in agreement with the integral relation. This residue calculation is a first-order consistency check; the four-chain argument, not equality of four derivatives, establishes the homology relation. To identify the embedding in the large-volume charge lattice, one must also fix the electric periods and their normalization. We do this next on a marked small-bulk neighborhood.

C.4 Electric continuation and exact flat coordinates

Denote (5.2) by Σ and keep $(v, a, b) = (z_2, z_3, z_4)$ sufficiently small and nonzero. At fixed spectator degrees (ℓ, m, n) , write the summation degree in (C.3) as (i, ℓ, m, n, j, k) . The ten denominator arguments, with the constant 1 omitted, are

$$(k_0, \dots, k_9) = (n, m, j - i - n, \ell + j - k, \ell - i, i, \\ k - j, i + 2m - j, k - \ell - 2m, m + n - \ell - k), \quad (\text{C.25})$$

and $k_* = 2m + n$. Two negative arguments make the scalar and every first derivative vanish. Requiring all but at most one argument to be nonnegative gives

$$\begin{aligned} 0 &\leq i \leq \max(\ell, m - \ell), \\ 0 &\leq j \leq \max(\ell + 2m, m + n - \ell), \\ 0 &\leq k \leq \max(2\ell + 2m, m + n - \ell). \end{aligned} \quad (\text{C.26})$$

These bounds follow by choosing the possible negative argument in (C.25) and solving the remaining linear inequalities. For example, if $i > \ell$, then $\ell - i$ is the only negative argument,

so $i + n \leq j \leq k \leq m + n - \ell$ and hence $i \leq m - \ell$. Thus each coefficient in (v, a, b) contains only finitely many local degrees, all of which can be summed exactly.

This regrouping also proves convergence beyond $|z_1|, |z_5|, |z_6| < 1$. Put $d = \ell + m + n$. The bounds imply $i + j + k \leq 5d$, $|k_i| \leq 3d$ and $k_* \leq 2d$. With one negative argument $-s$, the positive arguments sum to $k_* + s \leq 5d$; the nonzero derivative coefficient is a bounded charge-matrix entry times $k_*(s-1)!/\prod_{k_i \geq 0} k_i!$. The multinomial bound therefore bounds this coefficient by a constant times 10^{5d} . With no negative argument the same bound holds, up to polynomial factors from harmonic numbers. There are only polynomially many local summands at each d . For any fixed $M > 1$, all seven series $W_0, C_1, \dots, C_6$ consequently converge normally on $|z_1|, |z_5|, |z_6| \leq M$ when $|v|, |a|, |b|$ are sufficiently small. At $(v, a, b) = 0$, they are $W_0 = 1$ and $C_a = 0$. In a smaller neighborhood W_0 is nonzero.

The overlap with the original large-volume series fixes their continuation. Choose logarithms by continuation from positive small z_a , without coordinate winding. Then all seven electric periods

$$\Pi_0 = W_0, \quad \Pi_a = \frac{W_0 \log z_a + C_a}{2\pi i} = W_0 t_a \quad (1 \leq a \leq 6) \quad (\text{C.27})$$

are holomorphic and single-valued near Σ at nonzero small spectator parameters. This is analytic continuation of the marked periods, not substitution into a finite total-degree expansion.

The exact flat-coordinate equations. To evaluate the three local first derivatives on Σ , put

$$B_N(t) = \frac{1}{\Gamma(1+t)\Gamma(1+N-t)}. \quad (\text{C.28})$$

Apart from $(2m+n)!/(m!n!)$, the fixed-spectator coefficient factors into four such kernels. Their orders, integer indices and shifts under (ρ_1, ρ_5, ρ_6) are

N_α	I_α	ϵ_α	$\sum_\alpha N_\alpha = m,$
ℓ	i	ρ_1	$\sum_\alpha I_\alpha = m - \ell,$ $\sum_\alpha \epsilon_\alpha = 0.$
$2m - n$	$j - i - n$	$\rho_5 - \rho_1,$	
ℓ	$k - j$	$\rho_6 - \rho_5$	
$n - m - 2\ell$	$m + n - \ell - k$	$-\rho_6$	

(C.29)

The first-order sum may be extended to all integer i, j, k : outside the nonnegative orthant there are at least two negative arguments. Indeed, $i < 0$ would require both $j \geq 2m$ and $j \leq i + 2m$ if every other argument were nonnegative. With $i \geq 0$, $j < 0$ similarly requires $j \geq 2m$; with $i, j \geq 0$, $k < 0$ makes both $k - j$ and $k - \ell - 2m$ negative. The bounds above show that the first-order convolution has finite support, so no regularization of an infinite sum is involved.

For $N \geq 0$, the sequence $B_N(k)$ at integer k is $\binom{N}{k}/N!$, supported on $0 \leq k \leq N$. The Gamma recurrence gives

$$B_N(t) = \frac{B_{N-1}(t) + B_{N-1}(t-1)}{N}, \quad B'_N = B_N * B'_0, \quad B'_0(k) = \begin{cases} (-1)^k/k, & k \neq 0, \\ 0, & k = 0, \end{cases} \quad (\text{C.30})$$

where the recurrence is for $N \geq 1$, and the convolution identity is an identity of integer-indexed sequences. If all four orders are nonnegative, differentiating any one factor in their convolution gives the same result. The zero sum of the shifts in (C.29) therefore kills every local first derivative. If one order is $-s < 0$, then $B_{-s}(k) = 0$ for all integers and

$$B'_{-s}(k) = (-1)^{s+k} \prod_{r=1}^{s-1} (k+r). \quad (\text{C.31})$$

The other three kernels combine into a binomial kernel of degree $m+s \geq s$, which annihilates this alternating polynomial of degree $s-1$ by finite differences. If two orders are negative, the first derivative is already zero. These cases prove, to all orders in the spectator parameters,

$$C_1|_\Sigma = C_5|_\Sigma = C_6|_\Sigma = 0, \quad t_1|_\Sigma = t_5|_\Sigma = t_6|_\Sigma = 0. \quad (\text{C.32})$$

Normal convergence makes these analytic identities. Ordinary Vandermonde convolution also gives

$$[v^\ell a^m b^n] W_0|_\Sigma = \frac{(2m+n)!}{n!(\ell!)^3(m-\ell)!(2m-n)!(n-m-2\ell)!}, \quad (\text{C.33})$$

with a reciprocal negative factorial interpreted as zero. Finally, on Σ ,

$$2\pi i(t_2, t_3, t_4) = (\log v, \log a, \log b) + O(v, a, b). \quad (\text{C.34})$$

Their Jacobian with respect to the spectator logarithms tends to the identity, so these three flat coordinates are independent for sufficiently small nonzero bulk parameters.

The integral charges and their unit normalization. Let δ be any of the four vanishing spheres and let (p^0, P, q, q_0) be its charge in the continued large-volume frame. Picard–Lefschetz monodromy around its individual discriminant branch acts by

$$\Pi_\eta \longmapsto \Pi_\eta + \langle \eta, \delta \rangle \Pi_\delta. \quad (\text{C.35})$$

The period Π_δ is not the zero germ: its first derivative in the smoothing direction is nonzero. All seven electric periods (C.27) are single-valued, so each pairs trivially with δ . They span the electric Lagrangian sublattice in the large-volume marking; hence $p^0 = P = 0$. Dividing the vanishing period by W_0 and restricting to Σ then gives $q_2 t_2 + q_3 t_3 + q_4 t_4 - q_0 = 0$. Independence of the spectator flat coordinates forces $q_2 = q_3 = q_4 = q_0 = 0$. Thus every sphere lies in $\mathbb{Z}\beta_1 \oplus \mathbb{Z}\beta_5 \oplus \mathbb{Z}\beta_6$, with no magnetic or D0 component.

To fix its integer multiplicity as well as its direction, normalize the toric residue by $(2\pi i)^{-3}$, so that $W_0 \rightarrow 1$ at large volume. In the node model $uv + st = \mu$, the vanishing S^3 is a two-torus fibration over $q = uv \in [0, \mu]$. Integrating the residue gives $\pm(2\pi i)^2 \mu$ before this normalization, and hence $\pm\mu/(2\pi i)$ afterwards. The covectors (C.24) therefore have the same unit normalization as the limiting flat differentials $dt_a = d \log z_a / (2\pi i)$. For positive $(v, a, b) \rightarrow 0$, write $x_+ > 0 > x_-$ at each of the two values of r . With the indicated choice of orientations, the limiting covectors and the resulting exact charges are

r	x	(q_1, q_5, q_6)	q
b^{-1}	x_+	$(0, 1, 0)$	$C_1 = \beta_5$
b^{-1}	x_-	$(1, 0, 0)$	$C_2 = \beta_1$
0	x_-	$(1, 1, 1)$	$\iota_* e_1 = \beta_1 + \beta_5 + \beta_6$
0	x_+	$(0, 0, 1)$	$\iota_* e_2 = \beta_6$

(C.36)

The third row reverses the common residue orientation. The integer charges are locally constant, so their limiting values identify them exactly at nonzero small bulk parameters, not only at the boundary. Their span is saturated and their unique primitive relation is $\iota_* e_1 - \iota_* e_2 = C_1 + C_2$.

There is no homological torsion ambiguity in this identification. The vertices of the original fan polytope contain the standard basis of $\mathbb{Z}^4$. The polar polytope has the four vertices

$$(-1, 2, 1, -1), \quad (1, -1, -1, -1), \quad (0, -1, -1, 1), \quad (-1, -1, -1, 4), \quad (\text{C.37})$$

whose determinant has absolute value one. In both polytopes the vertices alone therefore generate the full lattice. The toric fundamental-group and Brauer-group formulas [54, Corollaries 1.9 and 3.9] give trivial fundamental and Brauer groups for the resolved threefold and its smooth mirror; in particular the mirror H_3 has no torsion.

The marking is fixed by the convergent continuation just described, on a nonempty small-bulk part of the four-node family. This proof does not compute transport to the particular point (C.20) or around arbitrary global loops. Further continuation must transport the charge frame as well as the periods. Nor does electric monodromy determine the regular parts of the magnetic periods or bound-state stability.

C.5 Regular magnetic periods and their Abel limit

We now determine the restricted regular terms. For $P = \sum_a p^a H_a$, put $\kappa_{Pbc} = PH_b H_c$, $g_b = C_b/W_0$ and

$$D_P = \frac{1}{2} \sum_{b,c} \kappa_{Pbc} \widehat{C}_{bc}, \quad \Psi_P = \frac{1}{4\pi^2} \left(\frac{D_P}{W_0} - \frac{1}{2} \sum_{b,c} \kappa_{Pbc} g_b g_c \right). \quad (\text{C.38})$$

This is the magnetic normalization of (C.5). With at least three negative denominator arguments, every second derivative vanishes. With two negative arguments k_i, k_j , its contraction

is

$$(PD_i D_j) \frac{(-1)^{-k_i - k_j - 2} (2m + n)! (-k_i - 1)! (-k_j - 1)!}{\prod_{r \neq i, j} k_r!}. \quad (\text{C.39})$$

All other arguments in this formula are nonnegative. The possible unbounded local-degree rays, at fixed (ℓ, m, n) , have negative pairs $(2, 4), (6, 7), (4, 9), (3, 9), (4, 6), (7, 9)$ in the zero-based column ordering of (C.2). The last two have $D_4 D_6 = D_7 E = 0$ on the hypersurface and contribute nothing. The first four have $D_i D_j = C_2, C_1, \iota_* e_1, \iota_* e_2$, respectively. Every other nonzero sector has $i, j, k \leq 4d$, where $d = \ell + m + n$; this follows from the remaining linear inequalities in (C.25).

The four unbounded sectors are one-dimensional after fixing their transverse indices. The following table gives all such indices for the unbounded integer h . An empty index interval contributes zero.

$D_i D_j$	(i, j, k)	transverse indices
C_2	(h, j, k)	$\ell + 2m \leq k \leq m + n - \ell, \quad k - \ell \leq j \leq k$
C_1	(i, h, k)	$0 \leq i \leq \ell, \quad \ell + 2m \leq k \leq m + n - \ell$
$\iota_* e_1$	$(h, h + p, h + p + q)$	$n \leq p \leq 2m, \quad 0 \leq q \leq \ell$
$\iota_* e_2$	(i, j, h)	$0 \leq i \leq \ell, \quad i + n \leq j \leq i + 2m$

(C.40)

Their respective lower bounds are

$$\begin{aligned}
h_{C_2} &= \max(\ell + 1, j - n + 1, j - 2m), \\
h_{C_1} &= \max(k + 1, i + 2m + 1, i + n), \\
h_{e_1} &= \max(\ell + 1, m + n - \ell - p - q + 1, \ell + 2m - p - q), \\
h_{e_2} &= \max(\ell + j + 1, m + n - \ell + 1, \ell + 2m).
\end{aligned} \quad (\text{C.41})$$

Take a common cutoff $h_0 = 8d + 2$, exceeding all these lower bounds. Below it the coefficient sum is finite. Above it, sum the transverse indices before summing h . To display the resulting rational functions, define

$$N_- = n - m - 2\ell, \quad N_+ = 2m - n, \quad K = \frac{(2m + n)!}{m!n!}, \quad x^r = \prod_{j=0}^{r-1} (x - j). \quad (\text{C.42})$$

The two nodal tails are zero when $N_- < 0$, and the two exceptional-curve tails are zero when $N_+ < 0$. Otherwise write the scalar tail for charge γ as $A_\gamma R_\gamma(h)$, where

$$\begin{aligned}
A_{C_2} &= \frac{K(-1)^{n+\ell} (n - 2m - 1)^{\ell + N_-}}{\ell! N_-!}, \\
A_{C_1} &= \frac{K(-1)^\ell (-\ell - 1)^{N_-} (n - 2m - 1)^\ell}{\ell! N_-!}, \\
A_{e_1} &= \frac{K(-1)^{m+\ell+N_+} (m + 2\ell - n - 1)^{\ell + N_+}}{\ell! N_+!}, \\
A_{e_2} &= \frac{K(-1)^m (-\ell - 1)^{\ell + N_+}}{\ell! N_+!},
\end{aligned} \quad (\text{C.43})$$

and

$$\begin{aligned}
R_{C_2}(h) &= R_{e_1}(h) = \frac{\Gamma(h-\ell)\Gamma(h+\ell-m)}{\Gamma(h+1)^2}, \\
R_{C_1}(h) &= \frac{\Gamma(h-m-n+\ell)\Gamma(h-\ell-2m)}{\Gamma(h-2m+1)\Gamma(h-n+1)}, \\
R_{e_2}(h) &= \frac{\Gamma(h-2m-2\ell)\Gamma(h-m-n+\ell)}{\Gamma(h-n+1)\Gamma(h-\ell-2m+1)}.
\end{aligned} \tag{C.44}$$

These formulas follow by the finite-difference identity

$$\sum_{j=0}^r (-1)^j \binom{r}{j} \frac{\Gamma(x-j+\lambda)}{\Gamma(x-j+1)} = (\lambda-1)^r \frac{\Gamma(x-r+\lambda)}{\Gamma(x+1)}, \tag{C.45}$$

and its forward version, applied to the factorial pairs in (C.40). The identity itself follows inductively from the Gamma recurrence.

Every nonzero rational tail is $O(h^{-m-2})$. Its poles are integers below h_0 , of order at most two. Partial fractions therefore evaluate its sum exactly in $\mathbb{Q} + \mathbb{Q}\zeta(2)$: the simple-pole residues sum to zero, giving rational harmonic number differences, while double poles give $\zeta(2) - \sum_{j=1}^{h_0-r-1} j^{-2}$ for a pole at r . Together with the finite terms, this determines every $[v^\ell a^m b^n]D_P|_\Sigma$ without truncating local degrees.

Convergence and the endpoint limit. There are constants c_0, c_1 independent of (ℓ, m, n) such that the absolute value of each fully summed coefficient is at most $c_0 c_1^d$. For the finite part, all Gamma arguments are $O(d)$, and the factorial multinomial bound used above applies also to (C.39); harmonic numbers and the number of terms add only polynomial factors. For the tail, all shifts in (C.44) have absolute value at most $4d+1$. Comparing each linear factor at $h \geq h_0$ with its value at h_0 gives, for a constant c ,

$$|A_\gamma R_\gamma(h)| \leq c^d |A_\gamma R_\gamma(h_0)| (h_0/h)^{m+2}. \tag{C.46}$$

The value at h_0 is bounded by the finite transverse factorial sum. Summing the last bound costs only $O(h_0)$ and another exponential factor in d . This proves normal convergence of the restricted bulk series.

To check that the sum is the continued period value, rather than an unjustified specialization, first put $z_1 = z_5 = z_6 = \rho < 1$. Each line in (C.40) carries its full weight ρ^{i+j+k} , and its h sum converges. In a transverse finite difference, replacing a difference by the extra weight costs a factor $O(1-\rho)$ and loses one inverse power of h . Consequently the difference from the endpoint tail is a finite sum of terms bounded, at fixed spectator degree, by

$$(1-\rho)^s \sum_{h \geq h_0} \rho^{ch} h^{-m-2+s}, \quad s \geq 1, \quad c \in \{1, 3\}. \tag{C.47}$$

Here is a uniform version of this estimate. If E_h is the unit shift in a transverse difference, the weighted difference satisfies

$$(1 - uE_h)^r = \sum_{s=0}^r \binom{r}{s} (1-u)^s u^{r-s} (1-E_h)^{r-s}. \tag{C.48}$$

The transverse orders in the four sectors have total at most $3d$; their weights u are fixed positive powers of ρ . For $1/2 \leq \rho < 1$, all bounded shifts of the residual weight cost only an exponential factor in d . Applying the Gamma recurrence to each remaining difference, and comparing its linear factors at $h_0 = 8d + 2$, bounds its contribution by

$$C^d \epsilon^s \sum_{h \geq h_0} \rho^{ch} (h/h_0)^p, \quad \epsilon = 1 - \rho, \quad p = s - m - 2, \quad 0 \leq s \leq 3d. \quad (\text{C.49})$$

Polynomial factors in d are absorbed in C^d (the degree-zero case is immediate). The coefficient at h_0 is bounded by the same finite factorial sums used in (C.46); the binomial factors in (C.48) add at most 2^{3d} . For $p \geq 0$, comparison with an exponential moment bounds (C.49), after increasing C , by

$$C^d \frac{p!}{h_0^p} \epsilon^{s-p-1} = C^d \frac{p!}{h_0^p} \epsilon^{m+1}. \quad (\text{C.50})$$

Since $p \leq 3d < h_0$, $p!/h_0^p \leq 1$. For $p = -1$ the bound is $C^d h_0 \epsilon^s (1 + |\log \epsilon|)$; for $p \leq -2$ it is $C^d h_0 \epsilon^s$. Every term with $s \geq 1$ tends to zero and has an exponential-in- d bound uniform in ρ . The $s = 0$ term has the summable endpoint decay h^{-m-2} and converges by dominated convergence. The number of transverse terms is polynomial in d , so the same majorant applies to their sum. On a smaller spectator polydisc it is summable also in (ℓ, m, n) , justifying interchange of the Abel limit and the spectator series. Its overlap with the large-volume Frobenius series fixes the continuation marking. The magnetic periods have the usual transverse vanishing-period logarithms, but their restrictions to Σ are the analytic functions just obtained.

For completeness, the constant and linear coefficients of $(D_{H_1}, \dots, D_{H_6})|_\Sigma$ are

$$\begin{aligned} [1]D_H &= 2\zeta(2)(1, 0, 0, 0, 1, 1), \\ [v]D_H &= [a]D_H = (1, 0, 0, 0, 1, 2), \\ [b]D_H &= (0, 0, 0, 10, 10, 10). \end{aligned} \quad (\text{C.51})$$

All these coefficient sums are real for positive spectator parameters.

The four conifold constants. An independent boundary calculation checks the coefficient and sign. At spectator degree zero, all first Gamma derivatives vanish. The nonzero local contributions to the cubic Taylor coefficient $\kappa_{abc} \partial_a \partial_b \partial_c / 6$ lie on the four rays hC_1 , hC_2 , $h\iota_* e_1$ and $h\iota_* e_2$, with $h \geq 1$. The other two possible two-zero sectors have zero divisor intersection; every three-zero sector has zero triple intersection. On each active ray the two reciprocal Gamma zeros supply h^{-2} . Writing $H_h = \sum_{j=1}^h j^{-1}$, the remaining logarithmic derivative contracts to $-2(H_h - H_{h-1}) = -2/h$. Thus the four complete ray sums give

$$4 \sum_{h \geq 1} \frac{-2}{h^3} = -8\zeta(3) \quad (\text{C.52})$$

in the cubic Gamma coefficient, or $-4\zeta(3)/(2\pi i)^3$ in the instanton prepotential. These are the four $\text{Li}_3(1)$ contributions with unit genus-zero index. Their sum compensates the change

of the perturbative Euler term exactly. The boundary calculation checks the constant; the polarization and reality argument proves the full period identity.

D Fibration data and reproducibility

D.1 Integral E-string root images

Across the C_1 flop the cubic changes by $\kappa_{\text{fl}}(A, B, C) = \kappa(A, B, C) - (AC_1)(BC_1)(CC_1)$. Intersecting the flat-model sections with D_3, D_4 gives the following curve pairings in the primitive divisor basis of Section 3:

$$\begin{array}{c|cccccc}
& D_2 & D_5 & D_6 & D_7 & D_8 & E \\
\hline
o_3 = -C_1 & -1 & 0 & 1 & 1 & 0 & 0 \\
o_4 = C_2 & -1 & 1 & 0 & 1 & 0 & 0 \\
s_3 & 0 & 0 & 1 & 0 & 1 & -1 \\
s_4 & 0 & 1 & 0 & 0 & 1 & -1 \\
F_3 = F_4 & 1 & 0 & 0 & 0 & 0 & 1 \\
r_3 = r_4 & 0 & 0 & 0 & -1 & 1 & -2
\end{array} \tag{D.1}$$

The unimodular pairing in Appendix B makes equality of these rows an integral homology statement. In particular $s_4 - s_3 = C_1 + C_2$, whereas the common root $r_i = s_i - o_i - F_i$ is a different class. This distinction is needed when assigning the elliptic Kaluza–Klein charge.

D.2 An elliptic K3 in the common slice

The slice $m_4 + 2m_1 + m_2 = 0$ of the Newton polytope has 30 lattice points. In an integral basis its polar has vertices

$$(-4, -2, -1), (0, 1, 0), (1, 0, 1), (1, 0, 0), (-1, -1, 0), (0, 0, 1). \tag{D.2}$$

The two pieces of the subdivision meet in this reflexive three-dimensional slice. The resulting K3 equation can be written, in an affine base coordinate s , as

$$a_0 w^2 + (b_0 u^2 + b_1 uv + b_2 v^2)w + u(c_0 u^3 + c_1 u^2 v + c_2 uv^2 + c_3 v^3) = 0. \tag{D.3}$$

Here a_0, b_2 are nonzero constants, $\deg b_1 \leq 2$, $\deg b_0 \leq 4$, and $c_j \in s\mathbb{C}[s]_{\leq 7-2j}$. An exact rational member is specified by the following coefficient vectors, ordered by increasing power of s :

$$\begin{array}{c|l}
a_0 & (4) \\
b_2 & (2) \\
b_1 & (3, 1, 5) \\
b_0 & (7, 4, 4, 9, 5) \\
c_3/s & (7, 4) \\
c_2/s & (1, 8, 5, 6) \\
c_1/s & (2, 3, 1, 5, 9, 4) \\
c_0/s & (5, 9, 8, 6, 5, 1, 3, 8)
\end{array} \tag{D.4}$$

Completing the square produces a binary quartic with coefficients

$$\begin{aligned} A &= b_0^2 - 4a_0c_0, & B &= 2b_0b_1 - 4a_0c_1, \\ C &= b_1^2 + 2b_0b_2 - 4a_0c_2, & D &= 2b_1b_2 - 4a_0c_3, & E_4 &= b_2^2. \end{aligned} \quad (\text{D.5})$$

Writing E_4 for the quartic coefficient, to distinguish it from the divisor E , its invariants are

$$I = 12AE_4 - 3BD + C^2, \quad J_4 = 72ACE_4 + 9BCD - 27AD^2 - 27E_4B^2 - 2C^3. \quad (\text{D.6})$$

One normalization of the Jacobian has $f = -I/3$, $g = -J_4/27$. The polynomial $4I^3 - J_4^2$ has degree 24 and a double zero at $s = 0$. Its quotient by s^2 is squarefree and coprime to I, J_4 ; the fiber at infinity is smooth. Thus this member has precisely $I_2 + 22I_1$. This is an exact K3 calculation, not a construction of the heterotic bundles or of a semistable total degeneration.

D.3 Reproducibility and the role of finite checks

The source distribution contains exact-arithmetic calculations in `checks/`, with commands and dependencies documented in `checks/README.md`. They reconstruct the fan, Chow ring, primitive lattices, local smoothing topology, mirror critical scheme, Laurent mutation, Frobenius coefficients and fibration data printed here. The explicit global smoothing family is the one constructed in [8].

The period statements do not follow from a finite expansion. The coefficient identities, analytic continuation estimates and integral normalization used in their proofs appear in Section 5 and Appendix C. The hypermultiplet threshold check differentiates the singular prepotential, verifies the four unit symplectic shifts and their combined Smith invariants, and checks that all neutral charge projections vanish. It also verifies the algebraic reconstruction of the supergravity period matrix. These are consistency checks of the printed charge data, not independent proofs of its geometric marking. Exact low-degree series comparisons test the period formulas; they are not substituted for their all-orders arguments. Numerical central-charge evaluations and the separate magnetic-sheaf calculations are discussed in the companion note [16].

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